The Accountability of Missed Opportunities:
Electoral punishments from public infrastructure expansion.

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Working paper. Comments welcome.

Abstract

Can public goods expansion generate electoral backlash against incumbents? Do voters hold incumbents accountable for policies from which they derive no direct benefit? Conventional accountability models suggest that public goods provision rewards incumbents, yet the dynamics through which grievances translate into punishment remain both theoretically underdeveloped and empirically underexplored. This paper explores how infrastructure construction can provoke voter backlash in areas unaffected by new developments. Infrastructure provision has inherent characteristics that create particular scenarios for accountability. The visibility and salience of infrastructure, coupled with its differentiated, non-universal spatial impacts and slow expansion, set the stage for territorial competition across constituencies. When policymakers decide the location of new facilities, non-winners are confronted with a tangible counterfactual: the opportunity to enhance their own situation. The perception of having missed a chance to benefit—rather than material loss per se—is at the core of their grievance and becomes the main motivation to punish the incumbent responsible. Using a novel dataset on hospital construction in Spain (2002–2019) and municipality-level electoral results, I analyze how the allocation of 81 new hospitals affected support for regional incumbents. Compared with regions without hospital expansion, municipalities whose accessibility improved increased their support for the incumbent by 2.14 percentage points, whereas non-impacted municipalities reduced it by 1.85 points. Punishments manifest both as lower turnout (-2.26 pp) and as gains for opposition parties (+0.81 pp). Contrary to expectations, however, these punishments do not vary with prior accessibility. This study provides insights into nuanced electoral dynamics around public goods, emphasizing the critical role of unequal allocations in shaping electoral accountability.

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1 Introduction

On October 1996, 8,000 residents of Andújar (Jaén, Andalusia) —a small town with a population of 36 thousands and the third-largest municipality in the province— gathered outside the Andalusian Parliament to demand the construction of a new hospital for their region. The protests were sparked by the death of Isabel Corral, a pregnant woman who died while traveling to give birth at the hospital in Jaén, the provincial capital. Less than three years later, on May 1999, the Andújar Regional Hospital opened. But citizen demands do not always yield such swift outcomes. On December 1984, Écija (Seville, Andalusia)— then the province's fifth-largest town, with 35 thousands inhabitants—experienced a day of violent protests calling for the construction of a new hospital. Although the Andalusian government's 1981 healthcare plan had initially proposed building a hospital in Écija, the final decision placed the facility in Osuna, a much smaller town of 16,000 people that offered a more central location relative to other settlements in the province. Unlike Andújar, Écija had to wait more than 30 years, until April 2016, for its new hospital to open. These two cases highlight the challenges policymakers face in responding to citizen demands, particularly when investments in public infrastructure are both costly and long-term. The delay in fulfilling such demands stems, at least in part, from the complexity and budgetary constraints associated with these large-scale projects. As a result, the expansion of localized public goods with diffuse costs can lead to public discontent. These phenomena constitute the core questions I aim to adress in this paper on the relationship between public infrastructure provision and voter accountability.

In this paper, I am to answer a seemingly counterintuitive question. Can public goods expansion generate electoral backlash against incumbents? Under which circumstances do voters hold incumbents accountable for policies whose goal is improving overall accessibility? Classical accountability models suggest that voters reward the expansion of public goods, particularly when these investments provide new services or improve access to existing ones (Adiguzel et al., 2023; Artés et al., 2022; de la Calle and Orriols, 2010). While winners from expansion have typically take the frontstage, less is know about those outside the scope of expansion. This *Accountability of Missed Opportunities* theory, however, highlights the need to examine this largely overlooked group.

Thus, I advance the claim that the spatial allocation of public infrastructure is marked by features that make it especially likely to produce grievances among non-winners of expansion. Relative to public funds, infrastructure projects have differentiated, non-universal spatial impacts (Ichino et al., 2018) and tend to expand slowly and gradually over time. These features make them particularly salient and visible (Bickers and Stein, 1996). Their provision draws public attention not only within, but also beyond, the constituencies that benefit, as citizens observe and interpret government priorities through the location of physical projects. This visibility, with a zero-sum bias (Davidai, 2023; Foster, 1965), in how people perceive government resources—that is, the belief that gains in one place necessarily imply losses elsewhere—sets the stage for territorial competition. When new facilities are built, communities that are not impacted often perceive the outcome as a missed opportunity—one that delays their own development prospects. Drawing on counterfactual thinking (Byrne, 2016; Epstude and Roese, 2008), citizens mentally simulate the benefits they might have received had the infrastructure been allocated to their area. This sense of loss is the key source of grievance among non-winners of infrastructure provision. Crucially, the intensity of these grievances is not uniform: it

depends on how valuable the lost opportunity is perceived to be. Areas that already enjoy relatively good accessibility may see little reason to react, whereas in places where the new facility would have substantially improved access to essential services, the perceived cost of exclusion is much greater. In other words, the counterfactual imagined by non-winners—what could have been—carries more weight in contexts where the potential gains in service provision would have been most significant.

This paper contributes to the literature on accountability and public goods by shifting the focus from rewards to grievances. While existing studies often explore how citizens reward incumbents for the provision of public goods, considerably less attention has been paid to how voters might react negatively when benefits fail to reach all areas. I propose that reactions to public goods expansion are not only shaped by its beneficiaries but also by those who remain unaffected yet are aware of these improvements—what I refer to as the non-winners. Most models of accountability treat non-impacted areas as appropriate counterfactuals to evaluate the effects of public goods provision, implicitly assuming that policy interventions have no effect outside targeted areas. In contrast, I argue that unintended political effects are likely to emerge beyond the areas directly benefiting from such interventions. In this context, the absence of treatment may itself become politically meaningful—effectively a treatment in its own right. As a result, observed differences in electoral behavior between policy winners and non-winners may conflate two distinct dynamics: the presence of rewards and the presence of grievances. The argument I advance here is that electoral rewards from winners may coexist with punishments from non-winners, and that a fuller understanding of public reaction requires disaggregating the responses of each group.

A second contribution lies in highlighting the analytical advantages of focusing on the particularities of infrastructure. Unlike broader categories of government spending, infrastructure projects—such as hospitals, schools, or energy plants—are especially visible and politically salient (Bickers and Stein, 1996). They signal government priorities in concrete ways and carry an inherent spatial dimension (Golden and Min, 2013): they benefit nearby areas while remaining out of reach for others (Ichino et al., 2018). By focusing on infrastructure, this paper uncovers electoral dynamics that are harder to detect when analyzing overall spending priorities or budget allocations without a spatial component. While infrastructure offers a particularly clear lens for studying these dynamics, the mechanisms described here are not unique to it. Negative reactions from non-beneficiaries are also likely to arise in other policy domains where the spatial distribution of benefits is visible and politically charged. This includes, for instance, the allocation of regional development funds in multi-level governance settings, such as the European Union's Cohesion Policy.

I construct a dataset on public hospital construction in Spain between 2002 and 2019, focusing on the period after the devolution of healthcare in December 2001, when regional governments gained responsibility for hospital construction. Data is sourced from the *Catálogo Nacional de Hospitales*, which provides annual reports on hospitals in Spain, supplemented with key milestone dates obtained from official reports and media sources. Between 2002 and 2019, 81 new hospitals were built, compared to 213 existing hospitals in 2002. I complement the dataset on hospital construction with municipal-level electoral data from regional elections, sourced from the Spanish Electoral Archive (Pérez et al., 2021). The analysis focuses on how the construction of new hospitals affects voter support for the regional incumbent. I hypothesize that municipalities not directly impacted by new hospital projects are more likely to express dissatisfaction with

the regional government, particularly those with poorer accessibility, as they perceive a greater missed opportunity for improvement. Results from a First-Difference Estimator specification shows that while both rewards are punishments for the regional incumbent emerge in response to hospital construction, punishments are larger and more pervasive across models –associated with a 2.4 percentage point reduction in vote shares across elections– than rewards, which corresponde to a 1.3 percentage point increase. Nevertheless, contrary to the expectations derived from the heterogeneous hypothesis regarding the value of the counterfactual, larger differences are observed in areas that already had better pre-construction accessibility.

2 Winners, losers and non-winners from policy.

Research on distributive politics has demonstrated that voters typically reward targeted spending (Veiga and Veiga, 2013) and targeted policies (Adiguzel et al., 2023; Artés et al., 2022; de la Calle and Orriols, 2010; Harding, 2015). However, there are instances where such policies prove electorally unproductive (Imai et al., 2020; Larcinese et al., 2013). The classical accountability model for public goods expansion predicts electoral rewards from the beneficiaries of policy implementation. In this paper, I refer to the beneficiaries as *policy winners*. Conversely, a more recent strand of literature applies accountability models to the retrenchment of public goods (Fetzer, 2019). For instance, the closure of healthcare facilities in Denmark (Nyholt, 2023) and the UK (Dickson et al., 2024) elicits reactions from those negatively affected —that is, *policy losers*. Grievances also arise from public service relative deprivation (Cremaschi et al., 2022; De la Cuesta, 2022) with no need of retrenchment. In both cases, there is a clear connection between individuals' welfare/utility derived from public services provision and their voting behavior: beneficiaries reward material gains, while those who suffer material losses punish incumbents.

While these findings align with classical accountability models, they leave important questions unanswered: Can public goods expansion provoke grievances in ways that classical models fail to predict? If so, under what circumstances might voters react negatively to policies that do not directly harm them? This paper advances the idea that public goods expansion may generate grievances when benefits are spatially constrained, particularly among voters who are aware of them but do not experience their effects. These grievances later translate into electoral punishments. Compared to benefits and electoral rewards, grievances and punishments remain both theoretically underdeveloped and empirically underexplored, particularly in the context of public goods expansion. To address this gap, I argue that the reaction to public goods expansion is not solely driven by its *winners*, who experience direct material effects, but also by its *non-winners*, those who remain unaffected.

Contrary to the prevailing assumption that voters are *myopic* about policy decisions beyond their constituency boundaries, I argue that for salient policy decisions where responsibility is clearly attributed across government layers, the expansion of public goods can generate *spillover effects* that extend beyond recipient areas. Despite visibility being a key requirement for accountability (de la Calle and Orriols, 2010), citizens are often depicted as passive observers of policy outside their immediate surroundings. In this paper, however, they are instead conceived as active evaluators of their broader social and political consequences. While some studies incorporate distances to public services to measure the extent of their impact (Adiguzel et al., 2023; Cremaschi et al., 2022), they remain focused on winners. By contrast, this paper highlights the experiences of those often overlooked— grieved voters who witness the implementation of public

goods without directly benefiting from them. It examines how visibility and salience allow these citizens to develop grievances, even in the absence of clear material losses.

To challenge the assumption of myopic voters, one must consider how they engage with policy when analyzing different forms of government action. On the one hand, when government action is based on funds, it is reasonable to assume that voters are unable to track the exact distribution of resources across constituencies or compare their share to others. Therefore, they tend to focus solely on their own direct gains when evaluating the incumbent. However, when the focus shifts to infrastructure—such as healthcare facilities, schools, or energy production plants—the situation changes. Citizens become more aware of and interested in public goods provision beyond their constituencies. This shift occurs for several key reasons. First, as scholars suggest, infrastructure projects are much more visible to voters than abstract government spending (Bickers and Stein, 1996; Stein and Bickers, 1994). The clearest expression of government spending is often found in its policies (Golden and Min, 2013). Public services, in turn, connect citizens' fiscal efforts to the tangible outcomes of state action, reinforcing their role in societal welfare. Voters are more aware of who is responsible for building new infrastructure than they are about the allocation of specific funds, which is often intermingled with other governments' expenditures. Infrastructure projects are not only visible but also highly salient, as they shape the delivery of key public services, influencing how citizens experience government action. Moreover, infrastructure projects require geographically defined decisions—such as the location of new facilities—which inevitably result in disparities in access (Ichino et al., 2018), regardless of whether policymakers are motivated by equity considerations or electoral calculations. These differences in accessibility arise because infrastructure investment is costly, and policymakers cannot ensure uniform service provision across the entire territory. The limited provision of these projects makes them highly visible and salient to the electorate, especially when compared to more abstract fiscal policies[1]. Finally, the significant expenses involved mean that budget constraints slow the development of infrastructure, increasing public scrutiny over both where and when new projects are built. Therefore, the visibility and salience of infrastructure, coupled with their differentiated, non-universal spatial impacts and slow expansion, set the stage for territorial competition across constituencies.

3 The Accountability of Missed Opportunities.

For this paper, I consider hospital network expansion in Spain as a well-suited example. Healthcare is a highly salient policy for citizens, consistently ranked among the top voter concerns in public opinion surveys (e.g., 30% of respondents identify it as a top issue (de Salud, 2024)). Moreover, hospitals are among the most frequently used public services, with approximately 30% of the Spanish population visiting a public hospital emergency room annually and 10% requiring hospitalization (de Salud, 2024). In Spain, the responsibility for healthcare infrastructure lies with the *Comunidades Autónomas*, which oversee the provision of medical services within their territories. Ensuring access to essential healthcare requires a network of hospitals, meaning that policy decisions extend beyond funding allocations or regulatory frameworks to include the physical infrastructure necessary for service delivery. Hospitals are not interchangeable with other forms of healthcare spending; they are fixed facilities that determine where and

[1]From a supply-side perspective, if infrastructure projects were not salient, policymakers would have little incentive to pursue their construction, given their high costs.

how citizens can access critical medical care. However, building new hospitals is costly, and budget constraints typically allow for only a limited number of facilities to be constructed at a time. Given this scarcity, policymakers must decide where to locate new hospitals among a range of potential sites. This decision is often made with considerable discretion, as multiple locations could reasonably justify investment based on demand and existing service provision. Since access to healthcare depends on physical proximity, the location of new hospitals creates disparities in service provision. Some areas gain direct access to a facility, while others remain dependent on more distant hospitals, reinforcing inequalities in healthcare accessibility. These disparities emerge not necessarily from explicit exclusion but from the inherent limitations of infrastructure expansion, where governments cannot provide a hospital for every community. This framework generalizes to other policy areas where infrastructure decisions shape access to public services, as education (schools) and transportation (train stations).

Figure 1 presents a simple model that guides this theory's argument. The subsequent images represent the geographical borders of a given constituency under which a government has competence over healthcare policy[2]. For simplicity, this political entity is depicted as a rectangle. At time t_0, there is a given distribution of hospitals —two facilities[3]–, labeled as H_0 and $H_{0'}$ (Figure 1a). Accessibility is determined by distance to hospitals. Therefore, the reference hospital for each location in the region is identified by its closest facility, and accessibility is measured as the distance to this reference hospital, $Ac = dist_{closest}$. Hospital catchment areas emerge by grouping locations that share the same reference hospital.

In this model, the policymaker prioritizes accessibility improvements when determining the location of new facilities[4]. The decision follows an optimization process that maximizes accessibility, minimizing the average distance from both existing and newly constructed infrastructure. This problem may yield multiple solutions –meaning that different potential locations could provide similar accessibility gains, denoted as H_A, and H_B. However, budget constraints limit investment to a single new facility. The policymaker must therefore select one alternative, opting for H_A (Figure 1c). The addition of this new hospital establishes one new catchment area, reducing travel distances for locations in its vicinity while accessibility remains unchanged for the rest.

[2]I assume a uniform distribution of relevant variables for accessibility across the area, including age, sex, socio-economic status, and even geographical features, eliminating potential confounding factors related to and allowing the focus to remain on the spatial dynamics of hospital placement.

[3]Appendix A.1 follows the same reasoning process with the example of a political entity with only one hospital at t_0.

[4]The distributive politics literature suggests that policymakers are, at least partially, electorally motivated, targeting policy towards areas where it is most productive for gaining support. On the one hand, incumbents benefit their strongholds (Cox and McCubbins, 1986); on the other hand, swing constituencies are the ones to target (Dixit and Londregan, 1996; Lindbeck and Weibull, 1993). Consequently, location decisions may be endogenous to incumbents' performance. This issue requires further discussion, which will be addressed when explaining the underlying mechanism in the following paragraphs.

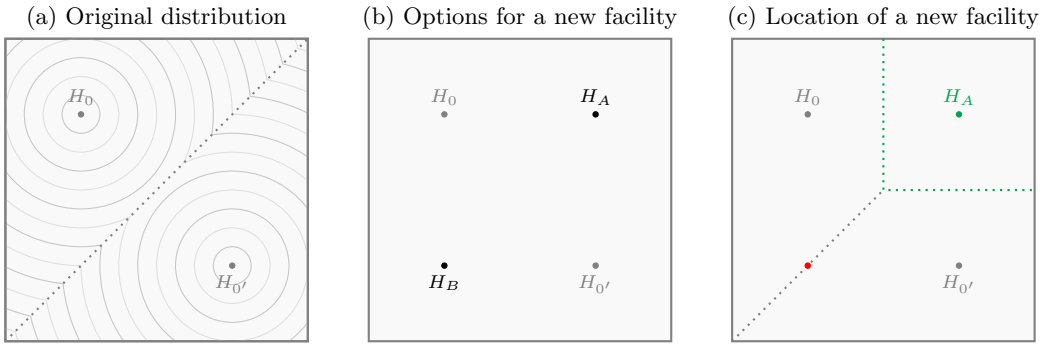


Figure 1: Model set up

This model represents a straightforward case of public goods expansion through the opening of a new hospital that provides services to citizens. It can be framed as a game in which policy-makers distribute an in-kind resource that cannot benefit all inhabitants in the constituency but only a portion of them. In short, there are direct winners but no explicit losers. Importantly, following classical accountability models, this process of deciding on the location of new facilities is not a zero-sum game. In a zero-sum scenario, where some level of redistribution occurs, resources would need to be reallocated from one place to another, creating both beneficiaries and those who experience direct losses, potentially leading to grievances. Here, however, non-winners simply maintain their existing conditions.

Figure 2 summarizes the model implications for accountability. The classical expectation of expansion (Figure 2a) (Adiguzel et al., 2023) entails that *winners*, those whose reference hospital shifted from H_0 or $H_{0'}$ to H_A, reward the government responsible of the policy (Figure 2a and Hypothesis 1).

Hypothesis 1 (Classical accountability of expansion)**.** *The expansion of public goods increases support for the incumbent from winners, those whose accessibility is improved.*

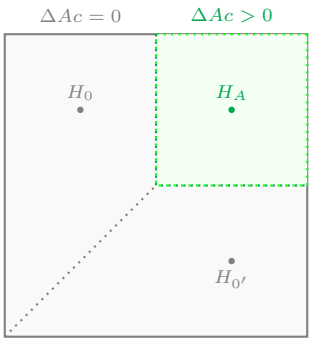


Figure 2: Implications for accountability

Challenging these assumptions, this paper builds on a paradox: voters feel aggrieved by hospital location decisions despite seeing no deterioration in their actual conditions. My central argument posits that voters perceive this process as a zero-sum game. Citizens interpret the expansion of public goods, such as healthcare infrastructure, as a contest for limited resources.

Social psychology and game theory scholars have shown that individuals often exhibit a zero-sum bias, in which they perceive resources in competitive terms: for someone to gain, someone else must lose (Foster, 1965). While this bias is particularly salient in conflicts between predefined groups—such as partisans (Roberts and Davidai, 2022), immigrants (Chinoy et al., 2023), or majority populations (Brown and Jacoby-Senghor, 2022)—its application to distributive policies remains underexplored[5]. Typically, zero-sum perceptions emerge in contexts where resources are not inherently scarce or even unlimited (Meegan, 2010). However, I extend this reasoning to infrastructure, which, by nature, is costly and slow to implement, reinforcing the perception that access to these investments is a competition in which only some can benefit.

In short, the expansion of infrastructure may create a perception of territorial competition, but grievances emerge not from direct competition, but from a sense of loss. When a new investment, such as a hospital, is allocated, non-winners realize that they have missed out on an opportunity that they valued. In this moment, a counterfactual thought arises. The literature shows that people often construct counterfactual alternatives to reality—mental representations of what might have been (Byrne, 2016; Epstude and Roese, 2008). In this paper, non-winners imagine a scenario in which they themselves were the recipients of this investment—where they are the winners, and the hospital is built closer to them. But in reality, they do not win. Instead, there is a sense of missed opportunity—the feeling that their own situation could have been improved, but was not. Non-winners compare their current, material situation not to the actual beneficiaries, but to an imagined world in which they are the winners. They feel they have missed the opportunity to enhance their own situation. This sense of loss, derived from the counterfactual thought, is the core of the grievance, even though their material situation has not changed. Benchmarking (Kayser and Peress, 2012) suggests that evaluations of policy actions are typically shaped by comparisons to the experiences of one's peers. In this context, however, voters do not compare themselves to the direct beneficiaries of the new hospital. Instead, their reference point is a counterfactual scenario where they themselves are the beneficiaries. This highlights an important distinction: the benchmarking process here involves comparing one's actual situation to a hypothetical outcome, rather than to the real, material experiences of others. Moreover, these grievances are particularly difficult to resolve. Hospital construction, like many infrastructure projects, unfolds over long time horizons, and citizens recognize that missing this opportunity means waiting —potentially for years—before another hospital is built closer to them. The slow implementation of infrastructure projects is a defining characteristic of these investments, and this temporal dimension amplifies the perceived stakes of each location decision. As a result, the sense that losing out now translates into a prolonged disadvantage in access to public goods is reinforced.

For this argument to hold, it is not necessary for voters to believe that the policymaker strategically placed hospitals to gain votes. In fact, the policymaker does not even need to be politically motivated. Given the inherent characteristics of hospital infrastructure—its high costs, slow implementation, and long-term planning—the policy may simply reflect a fair distribution aimed at maximizing accessibility. Even in such a case, voters may still perceive that they have lost the opportunity to have a hospital built closer to them. While experimental evidence on process fairness shows that voters punish incumbents when the allocation of public goods is politicized and visible (Balla et al., 2002; Bøggild, 2016; Tóth et al., 2022), the reaction to hospital location decisions does not require such explicit political targeting. Both politically

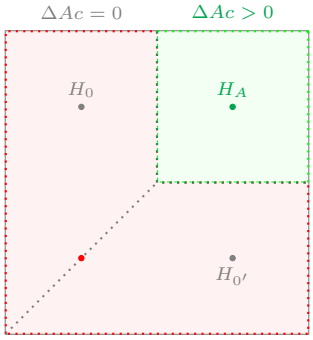
[5]For a review of other applications of zero-sum bias, see Davidai and Tepper (2023).

motivated and neutral hospital placements are expected to generate grievances. Even if policymakers could anticipate the electoral consequences of hospital siting—something that would entail a complex cost-benefit analysis unlikely to occur—their incentives to minimize negative reactions would not fully eliminate perceptions of loss. In this context, while punishments may be reduced when hospital placements aim to politically reduce grievances, they are unlikely to disappear entirely.

Figure 3 displays the implications for accountability my model offers: that non-winners—those who did not experience any accessibility improvement or deterioration—punish the incumbent government for the opportunity lost to improve, as shown in Figure 3a and Hypothesis 2A.

Hypothesis 2A (Accountability of missed opportunities)**.** *The expansion of public goods decreases support for the incumbent from non-winners, those whose accessibility remains constant but feel there is a missed opportunity for improvement.*

(a) H2A – Accountability of missed opportunities



(b) H2B – Accountability of missed opportunities: value of CF

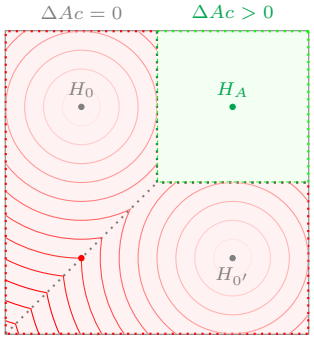


Figure 3: Implications for accountability

But are voters equally aggrieved when they miss the opportunity for new investments? The final stage of this paper's argument entail that, for some, the perceived loss is more acute because the counterfactual represents a significant improvement in their situation—an opportunity that would have meaningfully enhanced their access to healthcare. In contrast, for others, the counterfactual may only suggest marginal improvements, leading to a more tempered sense of loss. In this way, reactions will be stronger where baseline accessibility was low, and weaker in areas that already enjoyed close proximity to healthcare facilities. Referring back to Figure 3, this paper's theoretical model allows for the incorporation of different value assigned to counterfactuals, as Hypothesis 2B states. Figure 3b shows that grievances are conditional on the previous level of accessibility: in short, punishment from non-winners increases as voters' reference hospital becomes further away, signaling a larger missed opportunity.

Hypothesis 2B (Accountability of missed opportunities: value of counterfactual)**.** *The expansion of public goods decreases support for the incumbent from non-winners, particularly when the missed opportunity represents a significant improvement in accessibility.*

This theoretical framework resonates with the literatures on left-behind regions and geographies of discontent. However, these works leave unanswered questions about how patterns of resentment travel across different contexts. A growing body of research suggests that some places do not matter in economic and political decision-making (Rodríguez-Pose, 2018). Place-based

resentment arises when residents perceive that their communities are neglected by policymakers, for instance, when they feel they are not receiving their fair share of resources (Huijsmans, 2023). Place-based resentment builds through the accumulation of grievances stemming from systematic underprovision over time. In contrast, my argument does not exclusively apply to left-behind areas. While grievances are expected in underserved regions—where missed opportunities are likely to have a greater impact—territorial competition can also be felt in prosperous but overlooked areas that perceive the loss of an opportunity to improve. Moreover, geographies of discontent show that public service deprivation strengthens support for non-mainstream parties (De la Cuesta, 2022) by heightening perceptions of competition with migrants (Cremaschi et al., 2022). However, less attention has been paid to how specific policy interventions may generate grievances. While far-right support can ultimately emerge when grievances are linked to specific scapegoats—be they traditional parties or migrants—this process is not always immediate and requires certain political actors to shape and channel discontent. This paper aims to shed light on specific, short-term reactions in contexts where there were no challengers to mainstream parties, and accountability is straightforwardly shaped by electoral rewards or punishments for incumbents.

4 Hospital construction in Spain, 2002–2019

To test the hypotheses of this paper, I focus on the development of hospital infrastructure in Spain. It is relevant to note that healthcare policy in Spain is currently in the competence of Spanish regions[6]. Regional competence provides one fundamental advantage for the analyses in this paper. Regional authority provides a multiplicity of regional incumbents for political parties from different ideologies, in comparison to Spanish central governments, which, in the time span analyzed, had been only guided by two different parties. Thus, in this paper I only consider observations from municipalities once healthcare policy is fully granted to every Autonomous Community, that is after 2001.

Data on hospital locations are obtained through the digitalization of the Catálogo Nacional de Hospitales (CNH), the Spanish National Catalogue of Hospitals, an annual report that lists all hospitals operating as of December 31st each year. This catalogue provides a comprehensive inventory of public and private hospitals, including both general and specialty facilities[7]. I have compiled a comprehensive list of general hospitals with Emergency Rooms available for universal public use that were operational at any point between 2002 and 2019, covering all regions of Spain. I focus on general hospitals with emergency services open to the public, as these are the most likely reference point for voters when considering their nearest facility. These hospitals

[6]However, this decentralized model has not been in place since the establishment of Spain's democracy in the early 1980s. Healthcare had been gradually conferred to some Spanish regions since July 1981, when Catalonia acquired the competence, followed by Andalusia (1984), the Basque Country (1987), Comunitat Valenciana (1987), Galicia (1990), Navarre (1990) and Canary Island (1994). Finally, in December 2001, the Central Government decided to fully decentralize healthcare for the other ten regions that weren't granted it yet –except for Autonomous Cities of Ceuta and Melilla, where it retained the responsibility.

[7]Healthcare administration in Spain varies by region, despite overarching national guidelines. While most Autonomous Communities have fully publicly funded hospitals, some—such as Catalonia, the Basque Country, and Andalusia—rely as well on agreements with private providers to deliver public care. Additionally, there is significant variation in the medical services offered at each facility, ranging from large hospitals with comprehensive care units to specialized hospitals focused on psychiatric care, maternity, geriatrics, oncology, rehabilitation, ambulatory and short-stay treatments, or acute and long-term care.

serve the broadest share of the population and account for the largest volume of visits and medical care. For a limited number of cases, I include facilities that were not listed in the CNH but met the infrastructure criteria[8].

Instead of exploiting the original panel structure of the CNH, I have recorded key milestones for each hospital—such as the groundbreaking ceremony and the official inauguration—drawing on official records and media sources (Figure A.4.1 in Appendix A.4). These dates are central to the empirical strategy of the paper. I use the date of the groundbreaking as the key event marking the initiation of construction. This moment represents the first concrete step in transforming an electoral promise into a visible policy action, and it often receives public and political attention. Importantly, political responsibility for the project is attributed to the same incumbent who will face the next election. In contrast, inaugurations frequently occur under different governments—indeed, in nearly half of the cases in the sample, the hospital was opened by a government other than the one that initiated its construction[9]. While groundbreaking ceremonies may generate smaller electoral effects than the actual opening of a hospital, they serve as a clear and credible signal of forthcoming infrastructure and change in accessibility.

Hospital facilities embody the characteristics of infrastructures previously discussed. As previously stated, healthcare is a highly salient policy issue in Spain, and a significant share of the population makes use of public healthcare services each year. Although eligibility is universal, the benefits of new facilities are strongly conditioned by proximity: the expansion of hospital services is more valuable to residents living near the newly constructed building. At the same time, the number of hospitals built per province or region is limited. New hospitals entail substantial financial and administrative costs, and their construction is far from immediate. As illustrated anecdotally in this paper's introductory example, hospital construction often reveals a mismatch between demand and supply: while citizens frequently call for new hospitals, policymakers often face considerable obstacles in delivering them. In the sample used in this paper, the average time between groundbreaking and inauguration is 3.6 years, and 81% of hospitals are opened under a different legislature from the one that began construction [10]. Over the 18-year period analyzed, some Autonomous Communities (such as Navarre or Castilla y León) did not build any new hospitals. Still, a total of 81 new hospitals were constructed during this period, out of a total of 292 existing facilities. The case of hospital construction in Spain thus offers a valuable setting to explore how infrastructure provision may trigger internal territorial competition for public investment.

Figure 4 illustrates the hospital construction process from 2002 to 2019. The total number of hospitals increased significantly from 248 in 2002 to 292 in 2019. This expansion contributed to a reduction in the average distance between municipalities and their closest hospital, which decreased from 27.5 km to 26.2 km (Figure 4a). However, these aggregate numbers mask more nuanced trajectories of hospital development. Figure 4b offers a more detailed view of these trajectories. Among the 248 hospitals present in 2002, 213 remained operational throughout the entire period. By 2019, 81 new hospitals had opened, while 35 had closed. Yet, broad categories like openings and closures fail to capture the complexity of these changes. Specifically, 54 of the newly constructed hospitals provided services in areas previously without any hospital facilities.

[8]For example, some hospitals in rural areas are administratively attached to a reference hospital located in the provincial capital. As a result, although the catalog lists a single administrative unit, in practice, emergency services are provided in two different locations.

[9]See Appendix A.5 for a discussion on timing of hospital construction and a Electoral Infrastructure Cycle.

[10]See Appendix A.5

Additionally, there were 27 cases of relocation and renovation, where institutions moved to new buildings, typically improving their service offerings[11]. Nevertheless, the development of hospital infrastructure followed distinct regional patterns[12]. This expansion of hospital infrastructure in Spain contrasts sharply with the experience of other countries during the Great Recession[13].

Figure 5 represents accessibility changes between 2002 and 2019. A vast majority of Spanish municipalities, 76.28% of the sample, were not affected by new hospitals. In contrast, 15.09% of the population of municipalities improved their accessibility –that is, decrease their distance to the closest hospital. The maximum magnitude of this decrease is 64.6kms, 1% experienced larger than 31.2kms reduction, and 5% were bigger than 9.3kms[14].

4.1 Empirical strategy

Geographical variation in hospital impact and accessibility are examined through municipality-level (i) analysis, using regional election (t) results at this level as the outcome variable. While the theoretical framework refers to individuals behaviours, municipality-level election results provide the most consistent and informative unit of analysis, given the absence of panel data in Spain that tracks individuals' place of residence over time. Spain comprises over 8,000 municipalities, whose boundaries have remained largely stable[15]. This stability makes municipalities a suitable unit of analysis, offering a large number of observations and significant variation in accessibility across Spain's extensive territory. To capture this variation, I compile an annual dataset that includes municipality geometries and key controls, such as population socio-demographics, sourced from *Instituto Nacional de Estadística* (INE).

The electoral consequences of healthcare policy are analyzed at the regional level, as regional governments are responsible for these developments. Regional election results are sourced from the Spanish Electoral Archive (Pérez et al., 2021), which compiles electoral outcomes across different levels since the transition to democracy. While Spain's regional electoral systems differ in certain parameters—such as electoral thresholds or the unit of constituency delimitation—they all follow a proportional system with closed lists. However, the availability of municipality-level

[11] The numbers of stable hospitals and relocations do not always add up to the 2002 total due to cases where multiple hospitals merged into a single facility or where two smaller hospitals were replaced by a larger one. However, these changes had minimal impact on accessibility: while hospital mergers preserved existing facilities, substitutions and relocations generally occurred within the same municipality—with the largest recorded shift being just 5 km.

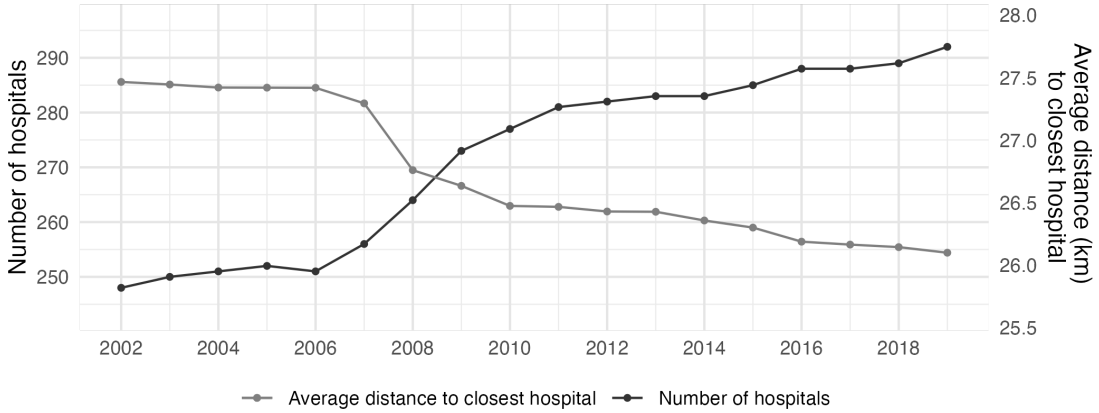
[12] Several regions, such as Navarra and La Rioja, did not build any new hospitals between 2002 and 2019. In other cases, hospital developments were limited to relocations or substitutions of existing facilities, as seen in Cantabria, Castilla y León, Galicia, and Asturias. However, the remaining eleven regions constructed at least one entirely new hospital in a previously unserved location. Figure A.2.1 in Appendix A.2 provides a detailed breakdown of hospital stock and development trajectories by region. Despite these variations in hospital expansion, improvements in accessibility have been modest or negligible in most regions. Notable exceptions include the Canary Islands, Madrid, the Balearic Islands, the Basque Country, and Andalusia, where new hospital developments have led to significant reductions in travel distance to healthcare facilities (Figure A.2.2 in Appendix A.2).

[13] In Denmark, for example, efficiency-driven austerity policies led to the closure of 30 out of 75 hospitals since 2007 (Nyholt, 2023).

[14] On the contrary, some municipalities suffered a deterioration of accessibility due to substitutions and relocations of existing hospitals, but this increment of distance is limited to less 4.6kms, what quantifies the maximum distance between the old facility and the new one. Overall, the extent and magnitude of improvements overshadow those of reductions. As relocated hospital are 5km apart at much, accessibility changes in nearby close municipalities are modest, while the largest variations were caused by new facilities in places with far hospitals.

[15] Exceptions, primarily due to mergers and splits, account for fewer than 50 cases.

(a) Stock of hospitals and accessibility



(b) Hospitals by trajectory since 2002

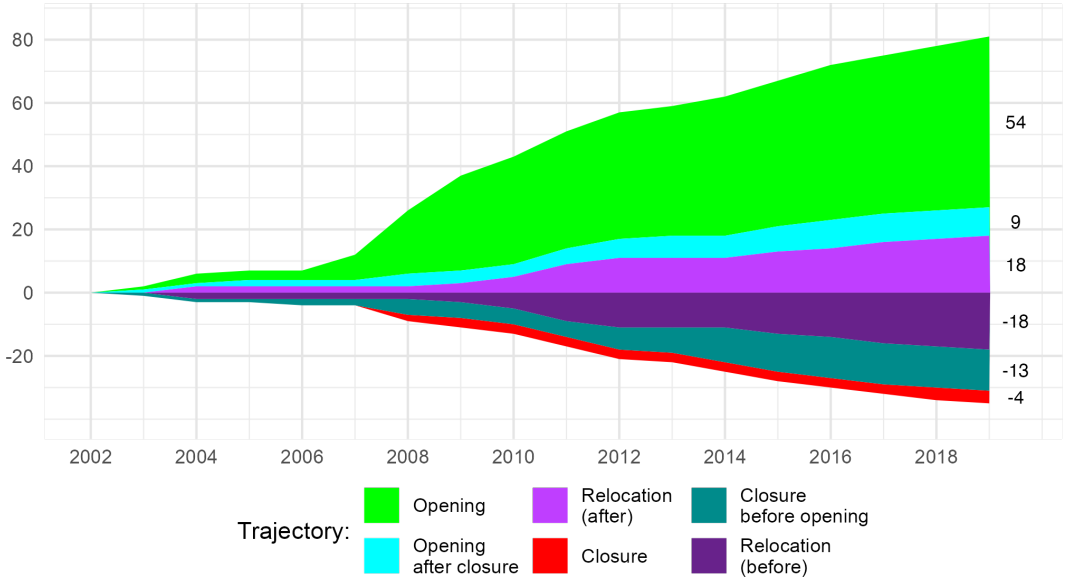


Figure 4: Hospital development in Spain, 2002-2019.

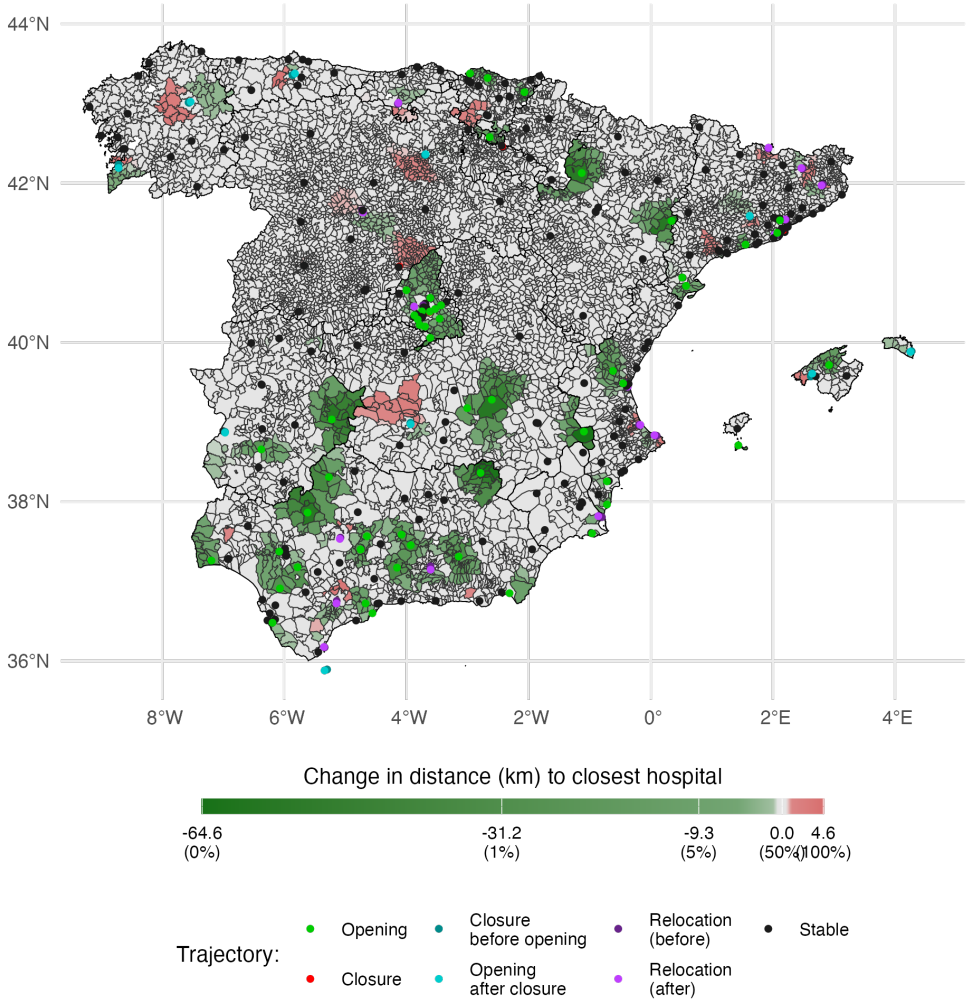


Figure 5: Hospital accessibility changes in Spain, 2002-2019.

election data varies across regions. Full and consistent results dating back to 2002 are available for 14 out of 17 regions[16], with partial data for three additional regions. Due to data inconsistencies, I exclude Castilla y León and the Canary Islands, while Ceuta and Melilla are also omitted from the analysis.

The hypotheses advanced in this article pose a methodological challenge regarding the choice of empirical strategies. The primary focus lies in examining the electoral behavior of areas that do not experience direct policy effects. This raises a central question: how can the political consequences of a policy be observed in places where no tangible change occurs—areas typically considered the counterfactual? To address this, the theoretical expectations are tested using a set of first-difference estimator models.

In this study, electoral accountability is assessed by examining whether voters reward or punish regional governments for initiating hospital construction projects. The dependent variable captures changes in electoral support for the regional executive responsible for the policy intervention. Specifically, I measure the difference in vote share (out of registered voters) received by the party or parties holding regional office between two consecutive elections: Y_{it}–$Y_{i(t-1)}$ = ΔY_{it}. Importantly, this is not simply the change in vote share for regional incumbents over time, but rather the change for each specific incumbent government in place at each election, recognizing that governing coalitions or parties may vary between elections[17]. To further unpack these dynamics, I replicate the models using alternative dependent variables: changes in turnout and changes in vote share for non-incumbent parties. This allows me to assess whether shifts in incumbent support reflect voter demobilization or electoral gains for the opposition.

To examine the relationship between hospital construction and electoral behavior, this paper operationalizes the impact of new hospital openings in two distinct groups: winners and non-winners. Winners are defined as those for whom the new facility becomes their reference hospital, while non-winners are those whose reference hospital remains unchanged. The main independent variable, $ImpNewHosp_{it}$, distinguishes between three categories that reflect changes in hospital accessibility since the previous election:

- *Improvement.* This category captures municipalities that are expected to experience improved accessibility because a new hospital, located closer to their current reference facility, began construction during the last legislature. These municipalities are considered *winners* (Hypothesis 1).

- *No impact.* This category captures municipalities that are not expected to experience any improvement in accessibility, as the new hospital that began construction during the last legislature is located further from their current reference facility. These municipalities are considered *non-winners* (Hypothesis 2A).

[16] Data for Aragón, the Canary Islands and Castilla y León are available but not yet included in these analyses.

[17] In particular, the analyses are conducted on two samples with slightly different outcome variables. First, I restrict the sample to legislative terms in which at least one hospital opening occurred, and compute the change in vote share for the party or parties responsible for the hospital construction. Second, I use the full sample of municipality-level elections and measure the change in vote share for the regional incumbent government. While these two outcomes are virtually identical—with only minor discrepancies due to cases in which the party responsible for hospital construction was removed from office through a vote of no confidence (*moción de censura*)—I employ both operationalizations as part of robustness checks, distinguishing between electoral reactions to parties responsible for initiating construction and those presiding over the actual inauguration, which, as noted earlier, often do not coincide.

- *No openings.* This condition includes municipalities from regions where no new hospitals are being built. It allows for the comparison of regions and terms with and without hospital expansion, as if a hospital would have not been built, while maintaining the integrity of the panel structure.

The categorical operationalization provides the average treatment effect of winners versus non-winners. While this paper primarily focuses on the electoral behavior of municipalities that are not benefited by the construction of new hospitals, those with positive direct accessibility gains differ in the extent of their benefits. Thus, impacts are conditional on distances[18]. Therefore, three alternative continuous operationalizations of the impact of new hospitals are proposed:

- $NewHosp_{it}$ measures the distance (log) from municipality i to the closest hospital under construction during the term within the region[19], serving as an indicator of proximity to policy investments. Although impacted municipalities are usually closer to the new hospital while those unaffected are further away, the distinction between winners and non-winners is blured and case dependent. As shown by Adiguzel et al. (2023), rewards should diminish the further citizens live from the new hospital. Additionally, I expect that as distance increases, rewards will turn into punishments.

- $Hosp_{it} - Hosp_{i(t-1)} = \Delta Hosp_{it}$ measures the change in distance (in kilometers) to the closest hospital across elections, serving as an indicator of absolute changes in accessibility. Non-impacted municipalities have a value of zero.

- $(Hosp_{it} - Hosp_{i(t-1)})/Hosp_{i(t-1)} = \Delta Hosp_{it}$ measures the change in distance (as a percentage of the previous term's distance) to the closest hospital across elections, serving as an indicator of relative changes in accessibility. As with the previous indicator, non-impacted municipalities have a value of zero.

By construction, the use of these three indicators entails the exclusion of observations from regions where no new hospitals are under construction during a specific term.

Finally, as suggested by Hypothesis 2B, punishments from non-winners are conditional on the significance of the missed opportunity, represented as a substantial improvement in accessibility. To measure the importance of these missed opportunities, I use $Hosp_{i(t-1)}$, which captures the distance from municipality i to the closest hospital within the region in the previous election, serving as a measure of prior hospital accessibility. Thus, within non-winner municipalities, punishments are expected to be larger the further away they were from their reference hospital, that is, at lower accessibility levels[20].

The model includes several control variables to account for demographic and economic differences across municipalities. Instead of total municipal population, I use the population living within a 10 km radius as a more accurate proxy for local settlement patterns and rurality than raw municipality population data. I also include the share of the population over 65, under 18, and of foreign origin to capture relevant sociodemographic structures. At the province level, unemployment rates are added to account for broader economic conditions that may influence electoral behavior.

[18] In this paper, distances are computed from hospital coordinates to municipality centroids.
[19] The closest newest facility in case more than one hospital has been built during the term.
[20] By symmetry, I expect that rewards increase as prior accessibility decreases.

The theoretical expectations presented in this article are tested using a set of first-difference estimator models. I adopt this specification to analyze both within- and between-unit effects. First-differencing enables the estimation of changes in the outcome variable—an approach that suits this paper, as the outcome must be manually computed to reflect the variation in vote shares of the parties that conform the regional executive in each election. Since incumbents may change between elections, this variation cannot be captured through standard fixed-effects estimation. Moreover, this specification accommodates both first-differenced independent variables—capturing temporal variation within units, such as accessibility changes measured by variation in distances to the closest hospital—and non-differenced covariates that explain cross-sectional variation, such as prior accessibility levels. In these models, I only consider observations from municipalities once healthcare policy is fully granted to every Autonomous Community, that is, after 2001[21]. I also include election fixed-effect[22]. The full model is exemplified in Equation 1:

$$\begin{aligned}\Delta Y_{it} =& \beta_1 ImpNewHosp_{it} + \beta_2 Hosp_{i(t-1)} + \\ &+ \beta_3 (ImpNewHosp_{it} \times Hosp_{i(t-1)}) + \\ &+ \beta \Delta X_{it} + \beta X_{it} + \gamma_t + \epsilon_{it}\end{aligned} \tag{1}$$

where ΔY_{it} refers to the change in the vote share of the parties that form the regional executive. $ImpNewHosp_{it}$ captures the exposure of municipality i to newly constructed hospitals during term t; this term can be alternatively operationalized using any of the continuous indicators described earlier (e.g., absolute or relative changes in distance). $Hosp_{i(t-1)}$ measures baseline accessibility prior to the election, expressed as the distance to the closest hospital in the previous term. The interaction term ($ImpNewHosp_{it} \times Hosp_{i(t-1)}$) tests whether the electoral impact of new hospitals depends on prior levels of accessibility. Control variables enter the model both as first-differences, ΔX_{it}, to account for within-unit variation, and in levels, X_{it}, to capture cross-sectional differences. Finally, γ_t denotes election fixed effects.

5 Results

Table 1 shows the effect of hospital construction—measured after the groundbreaking event—on the change in vote share across elections for the incumbent parties responsible for the project. The first two models restrict the sample to legislative terms in which at least one new hospital was initiated, while the last ones include the full sample of regional elections. Model 2 indicates that municipalities positively affected by a new hospital see an increase of at least 2.58 percentage points in the vote share for the competent regional incumbent, relative to municipalities that were not impacted. However, this result could reflect several dynamics: (i) a sizable reward

[21] I do not include pre-reform regional electoral results from regions that already had this policy conferred, in order to enhance consistency of regional variation across time.

[22] Regional elections are not consistently held across Spanish regions: while a group of regions are required to hold elections on fixed dates, others have discretionary power to decide their election day within a four-year window. Thus, I identify the first election day after December 2001 as the reference year (coded as 0) and code the rest of the election years accordingly. For regions with discretionary control over the electoral calendar, I assign the 0 to the election closest to December 2001, regardless of whether it occurred before or after that date. This approach is similar to a system of time dummies with a three- to four-year gap at the start of the time frame, while regional elections do not overlap in time: for instance, the latest fourth election in any region occurred before the earliest fifth election.

in line with classic accountability models of public goods provision, (ii) a sizable punishment from non-impacted municipalities, or (iii) a simultaneous combination of both effects. I present several pieces of evidence to help disentangle these mechanisms.

	Openings: Control + Impact	Openings: + Accessibility	All: Control + Impact	All: + Accessibility
Improvement	2.884***	2.577***	2.046***	2.144***
	(0.235)	(0.244)	(0.348)	(0.351)
No impact			-1.819***	-1.848***
			(0.148)	(0.148)
Distance closest hospital (log), t-1		0.521***		-0.220*
		(0.114)		(0.096)
Num.Obs.	6732	6732	21729	21729
R2	0.352	0.354	0.254	0.254
R2 Adj.	0.351	0.353	0.253	0.253
AIC	41214.6	41195.8	150249.0	150245.8
BIC	41323.6	41311.7	150392.8	150397.5
Log.Lik.	-20591.290	-20580.908	-75106.498	-75103.887
RMSE	5.15	5.15	7.67	7.67

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 1

First, Model 4 in Table 1 includes municipalities from regions and legislative terms without any hospital expansion, serving as a counterfactual. In this specification, coefficients capture the effect of being impacted or not by a new hospital, relative to municipalities in regions with no new hospitals at all. Here, municipalities that experienced improvements see a 2.14 percentage point increase in support for the incumbent compared to the no-expansion baseline, whereas non-impacted municipalities show slightly smaller decline—about 1.85 percentage points. Using the full sample highlights a broader dynamic: overall, punishments from aggrieved non-beneficiaries are comparable to rewards from those who benefit. However, considering that when a new facility is built, the share of municipalities experiencing improved accessibility is much smaller than the share that remains unaffected, an asymmetry of the average potential effect of new hospitals is striking given.

In terms of predicted values, Figure 6a displays the estimated outcomes for each category in Models 2 and 4. Both specifications show positive gains in incumbent party vote shares in municipalities impacted by hospital construction, alongside reductions in support in non-impacted municipalities. These non-impacted areas exhibit significantly different patterns not only compared to municipalities that experienced accessibility improvements, but also relative to those in regions where no hospital was built. Notably, the absence of hospital construction is associated with non-significant changes in vote share, suggesting that electoral reactions are specific to targeted infrastructure interventions.

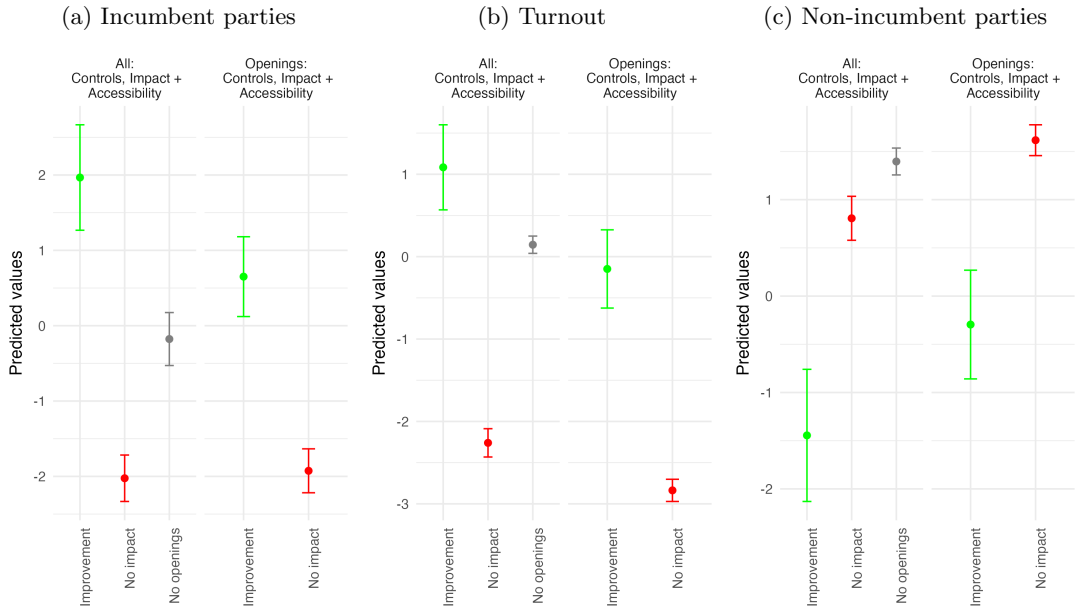


Figure 6: Rewards vs. Punishments: Predicted values

Reductions in vote share may result from two distinct voter reactions: grievances can either demobilize voters or lead them to support opposition parties. Figures 6b and 6c illustrate the presence of both dynamics. In impacted municipalities, rewards for the incumbent parties (1.97 pp) are accompanied by increases in turnout (1.08 pp) as well as reductions in vote share for non-incumbent parties (-1.44 pp). In contrast, in non-impacted municipalities, punishment of the incumbent (-2.02 pp) is associated with lower turnout (-2.26 pp) and gains for the opposition (0.81 pp), indicating that voters both abstain and shift their support away from governing parties.

These results are robust across a range of alternative specifications. Figure A.7.1 reflect similar results considering the inuaguration ceremony rather than groundbreaking event. When relocations of existing infrastructure are included as new hospitals in the analysis, the difference between impacted and non-impacted municipalities remains highly significant (see Figure A.7.2 in the Appendix). Similar patterns are observed when using alternative measures of accessibility. Whether accessibility improvements are measured using absolute distances in kilometers or standardized distances within regions—which account for geographical disparities across Spanish territories— rewards and punishments remain robust (Figure A.7.3). Furthermore, Figure A.7.4 in the Appendix presents results from models estimated without covariates or term fixed effects. Even under these more basic specifications, differences between winners and non-winners persist in most models. In both cases, the difference between the no-expansion baseline and impacted municipalities is not statistically significant.

These results suggest that both electoral rewards and punishments are simultaneously at play in shaping voter responses to hospital construction, supporting Hypotheses 1 and 2A. However, while the magnitude of the negative reaction from non-winners appears to offset the positive gains among winners, the prevalence of non-impacted areas relative to impacted areas may make punishments a more decisive force in explaining overall vote share changes.

Could these patterns be an artifact of a dichotomous operationalization? The previous models relied on a binary classification of impact: municipalities were either considered affected—if

the new hospital became their closest facility—or not. However, the degree to which a new hospital improves accessibility is inherently continuous. To account for variation in the intensity of impact, I introduce three alternative operationalizations: (i) the change in distance (in kilometers) to the closest hospital between elections; (ii) the same change expressed as a percentage of the initial distance to the nearest facility, capturing relative improvement in accessibility; and (iii) the distance to the newest hospital under construction.

Table 2 presents the results for these alternative operationalizations. Model 2 indicates that an expected decrease of 10 km in the distance to the closest hospital increases the regional government's vote share by 0.80 percentage points. In contrast, a 10% decrease in the distance to the closest hospital significantly increases the incumbent's vote share by 0.37 percentage points. Finally, a 100% increase in the distance to the hospital (i.e., the hospital being twice as far) is associated with a 0.76 percentage point decrease in the incumbent's vote share. These results are more clearly illustrated in Figure 7.

	Accessibility improvement (km) Openings: Control +	**Accessibility improvement (km) Openings: + Accessibility**	**Accessibility improvement (%) Openings: Control + Impact**	**Accessibility improvement (%) Openings: + Accessibility**	**Distance newest hospital Openings: Control + Impact**	**Distance newest hospital Openings: + Accessibility**
Change in accessibility (km)	-0.107***	-0.080***				
	(0.016)	(0.017)				
Change in accessibility (%)			-4.423***	-3.685***		
			(0.499)	(0.515)		
Distance newest hospital (log)					-1.175***	-1.103***
					(0.085)	(0.086)
Distance closest hospital (log), t-1		0.719***		0.648***		0.652***
		(0.114)		(0.114)		(0.111)
Num.Obs.	6732	6732	6732	6732	6732	6732
R2	0.342	0.346	0.345	0.349	0.356	0.359
R2 Adj.	0.341	0.344	0.344	0.347	0.355	0.358
AIC	41321.7	41284.2	41285.8	41255.6	41175.3	41142.6
BIC	41430.7	41400.1	41394.8	41371.5	41284.3	41258.4
Log.Lik.	-20644.849	-20625.102	-20626.908	-20610.801	-20571.634	-20554.289
RMSE	5.20	5.18	5.18	5.17	5.14	5.13

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 2

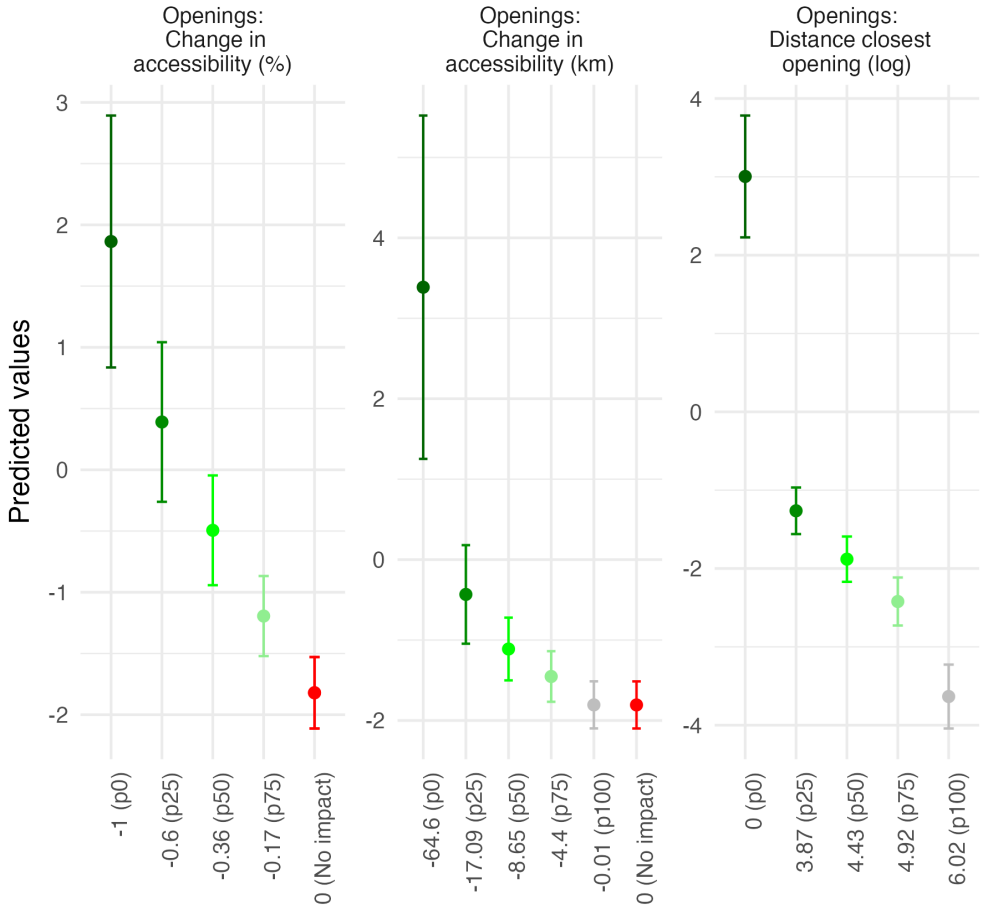


Figure 7: Alternative operationalizations of impact

Since visibility is a key requirement for holding policymakers accountable, it is plausible that those living farthest from a new hospital may be unaware of its existence. As a result, positive electoral rewards are expected in areas closer to the hospital, while voter dissatisfaction—or even punishment—may arise at intermediate distances. This implies that the effect of distance on electoral outcomes could be non-monotonic. Figure 8 presents the predicted values from linear, quadratic, and cubic models. All three specifications show a general downward trend as distance increases, with variation mainly at the extremes. Across models, electoral rewards are concentrated in municipalities located within approximately 12.8 to 20 km of the hospital. Beyond 24.5 km, these effects shift to statistically significant punishments. In the cubic model, the magnitude of punishment appears to decline after 148.4 km, becoming statistically insignificant beyond 403.4 km.

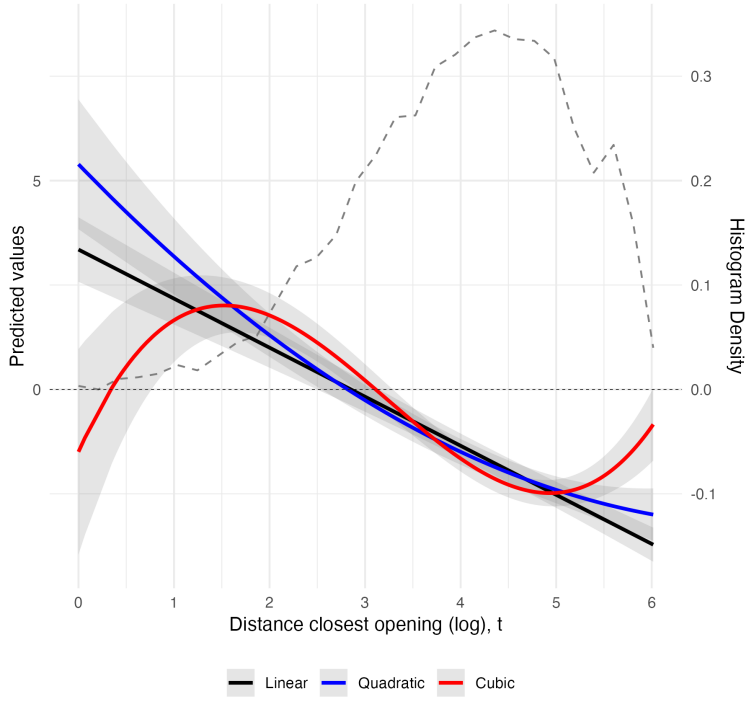


Figure 8

According to Hypothesis 2B, the expansion of public goods should decrease support for the incumbent among non-winners, particularly when the missed opportunity represents a significant improvement in accessibility. However, the results reveal no significant pattern (Figure 9). The conditional effect of the distance to the closest hospital in the previous election suggests that differences between improved and non-impacted municipalities are independent to distance to the closest hospital (Table 3).

	Openings: + Interaction	All: + Interaction
Improvement	2.354+	2.629
	(1.348)	(1.972)
No impact		-2.349***
		(0.437)
Distance closest hospital (log), t-1	0.520***	-0.276*
	(0.115)	(0.107)
Improvement x Distance closest hospital	0.069	-0.144
	(0.412)	(0.601)
No impact x Distance closest hospital		0.186
		(0.153)
Num.Obs.	6732	21729
R2	0.354	0.254
R2 Adj.	0.353	0.253
AIC	41197.8	150248.2
BIC	41320.5	150415.9
Log.Lik.	-20580.894	-75103.088
RMSE	5.15	7.67

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3

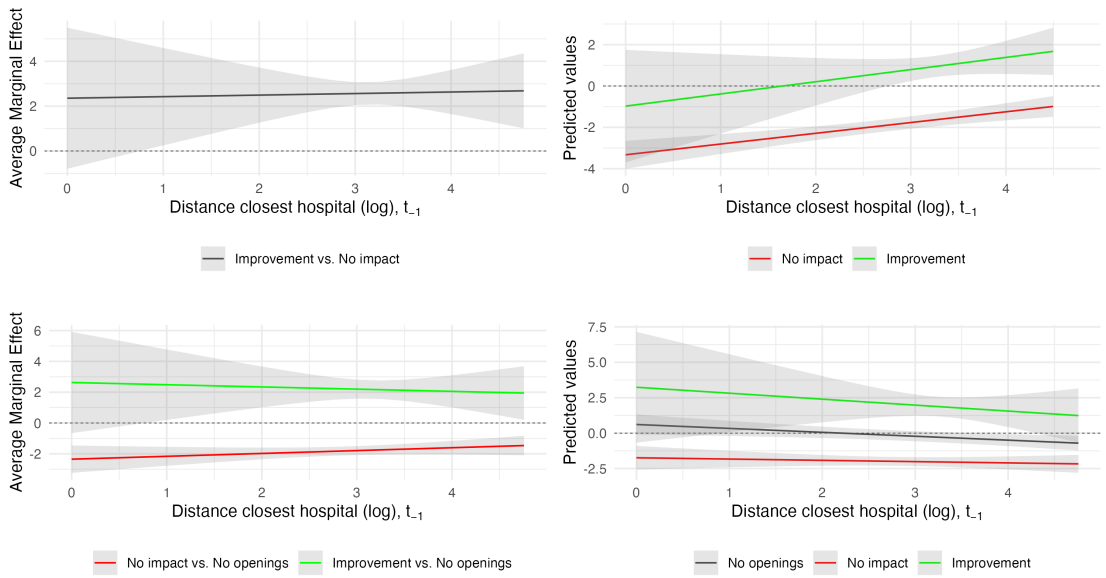


Figure 9

5.1 Expansion of hospital network in Andalucia, 2004

Work in progress.

6 Conclusions

This paper has examined whether the expansion of public infrastructure can generate electoral backlash among those who do not benefit from it. Drawing on a novel dataset of hospital construction in Spain between 2002 and 2019 and municipality-level results from regional elections, I find that electoral rewards and punishments coexist. Municipalities whose accessibility improved with a new hospital increased their support for the regional incumbent, whereas non-impacted municipalities reduced it, through both lower turnout and gains for opposition parties. Because non-impacted municipalities far outnumber impacted ones, punishments may weigh more heavily on aggregate vote changes than rewards. Contrary to expectations, however, the reaction of non-winners does not vary with their prior accessibility. These findings suggest that the absence of treatment can itself be politically meaningful, and that evaluations of distributive policies that treat non-beneficiaries as a neutral baseline may conflate rewards with grievances. Future work will exploit the Andalusian hospital expansion plan to strengthen causal identification.

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A Appendix

A.1 Accountability models for a one-hospital example.

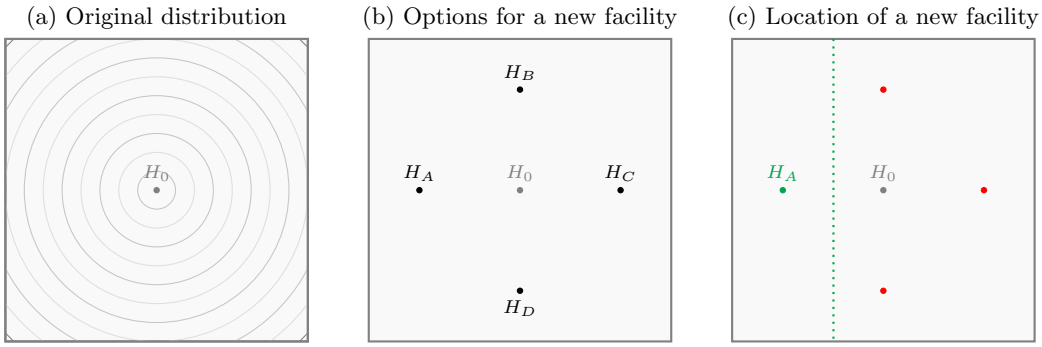


Figure A.1.1: Model set up

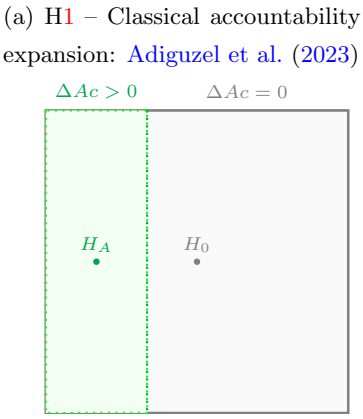


Figure A.1.2: Implications for accountability

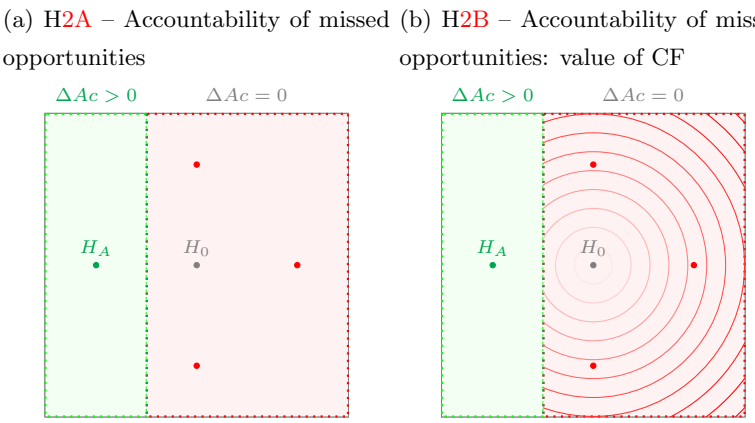


Figure A.1.3: Implications for accountability

A.2 Hospital development in Spain, 2002-2019, by region.

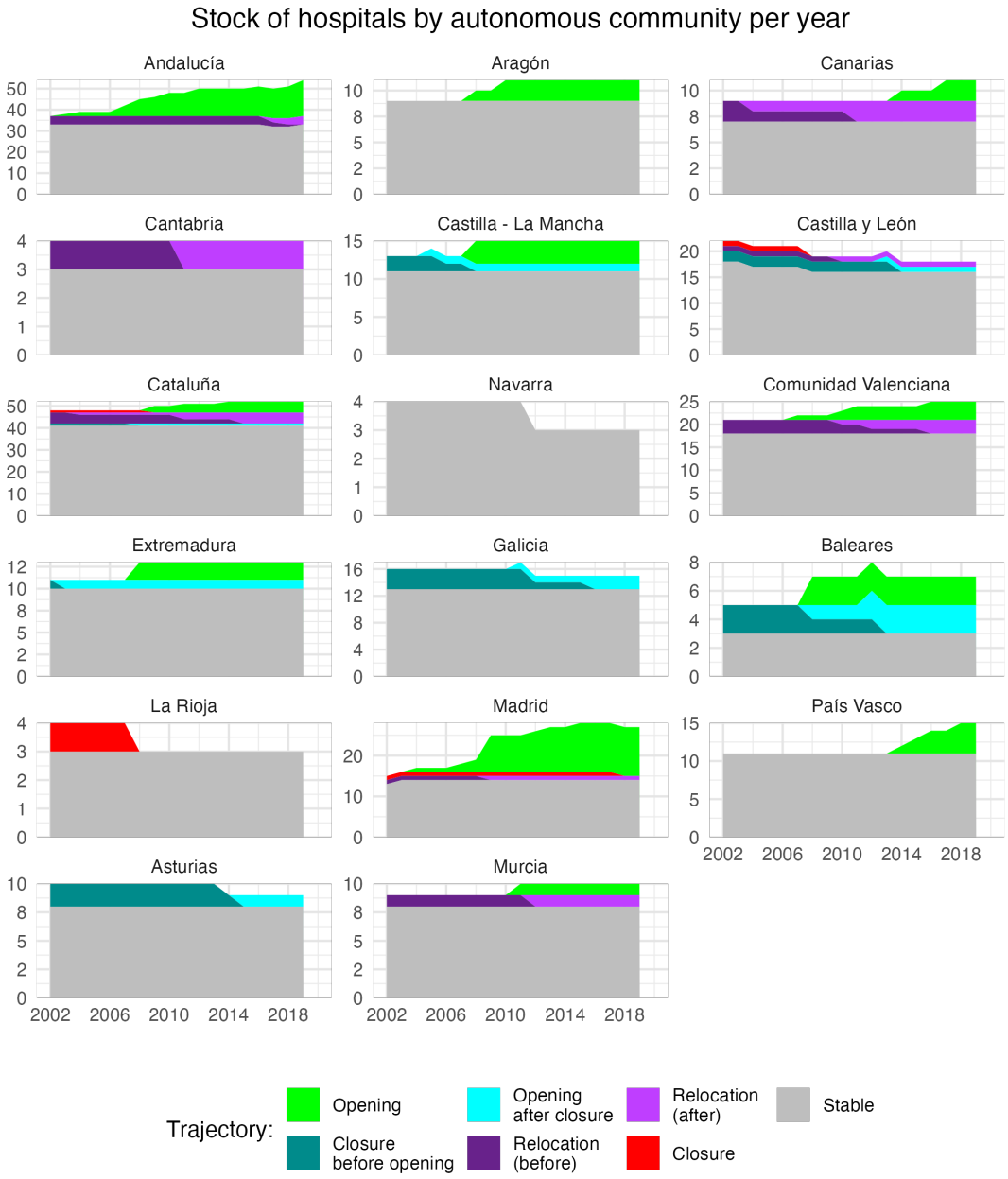


Figure A.2.1: Stock of hospitals by trajectory since 2002 and region

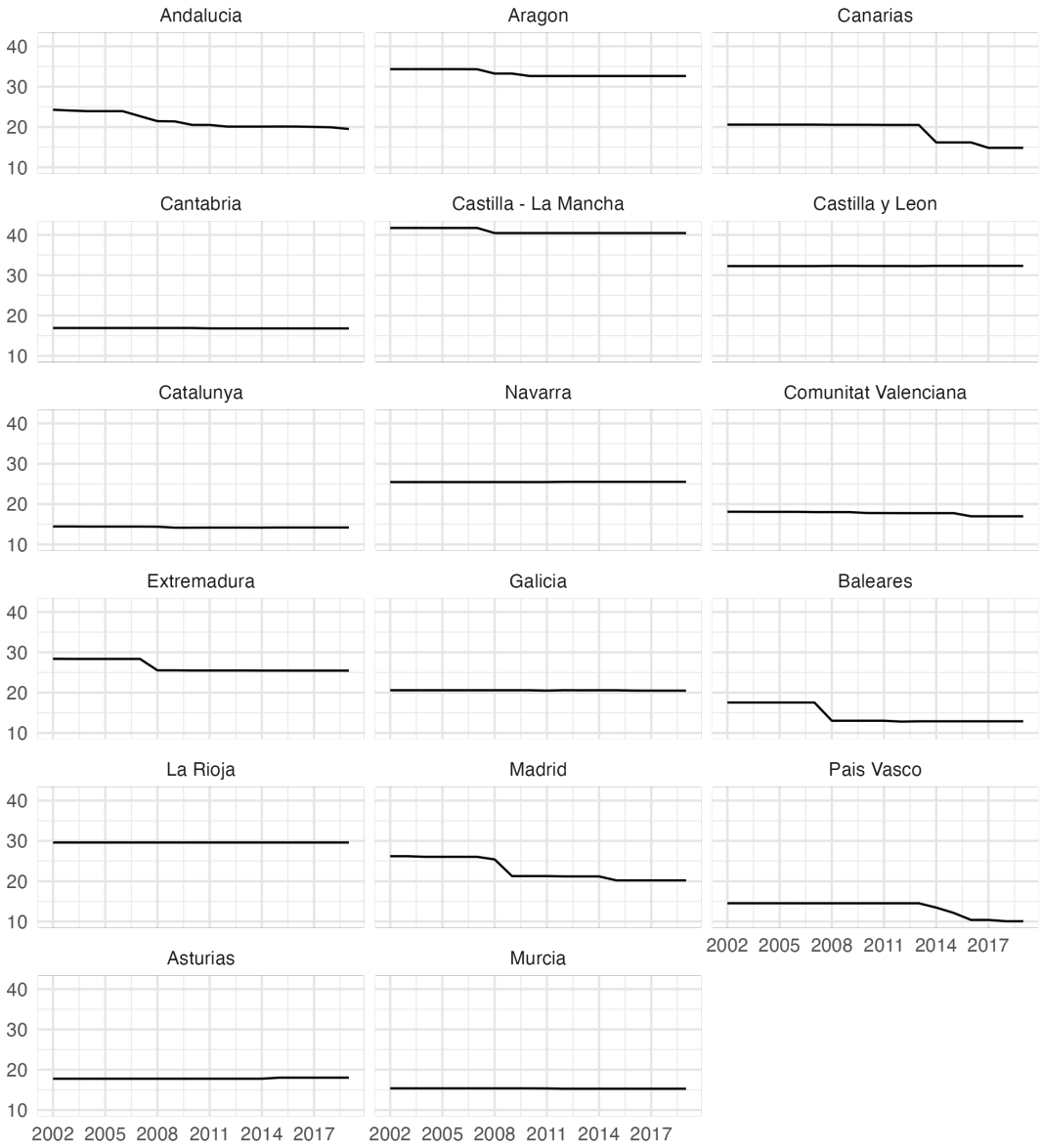
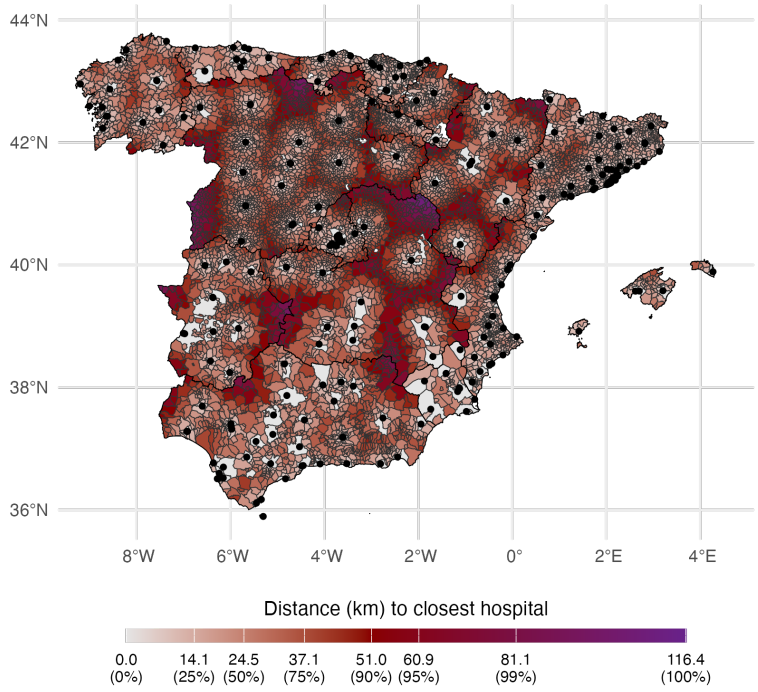


Figure A.2.2: Hospital accessibility by region

A.3 Hospital accessibility in Spain, 2002-2019.

(a) Accessibility in 2002



(b) Accessibility and hospitals by trajectory in 2019

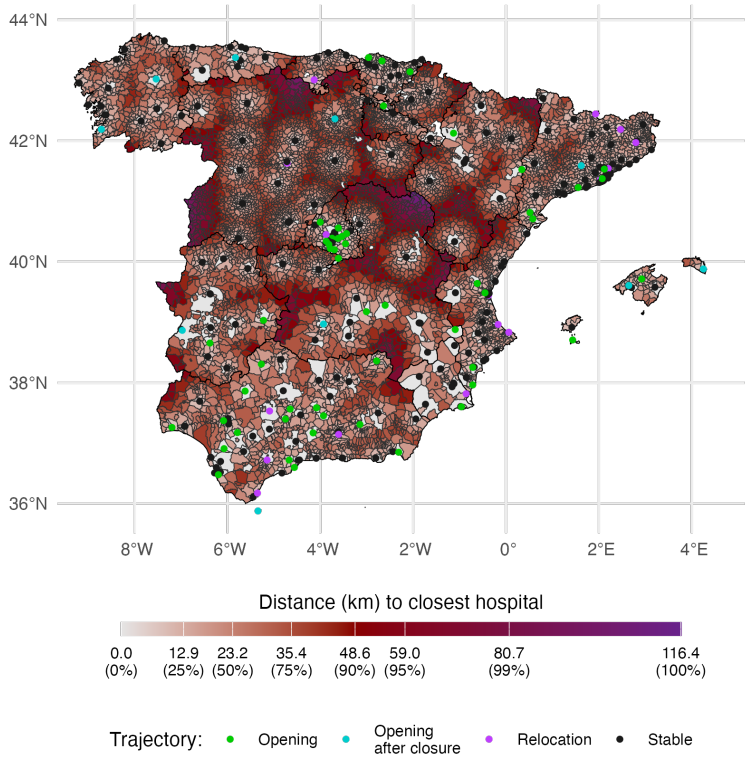


Figure A.3.1: Hospital accessibility in Spain, 2002-2019.

A.4 Timeline of hospital milestones and regional elections, 2001-2019.



Figure A.4.1: Timeline of regional elections, hospital milestones (groundbreaking and inauguration of new and relocated hospitals) and devolution of healthcare competences, by region, 2001-2019.

A.5 An Electoral Infrastructure Construction Cycle?

Politicians strategically manipulate policy around elections to increase their chances of re-election, often implementing popular measures before election.

- Most evidence focuses on the expenditure side: Election Budget Cycles, with increases in public spending before elections.
- Some recent (though modest) evidence on the revenue side, such as post-election unpopular tax reforms.

But these patterns are based on immediate implementation.

In my sample, only 19% of hospitals are inaugurated before elections take place. On average, there is a 3.6-year gap between the groundbreaking and the inauguration (with a median of 2.9 years). As a result, 81% of hospital projects could potentially be inherited by a different regional incumbent party. In practice, however, 51% of the total number of projects are ultimately inaugurated by the same party that initiated them.

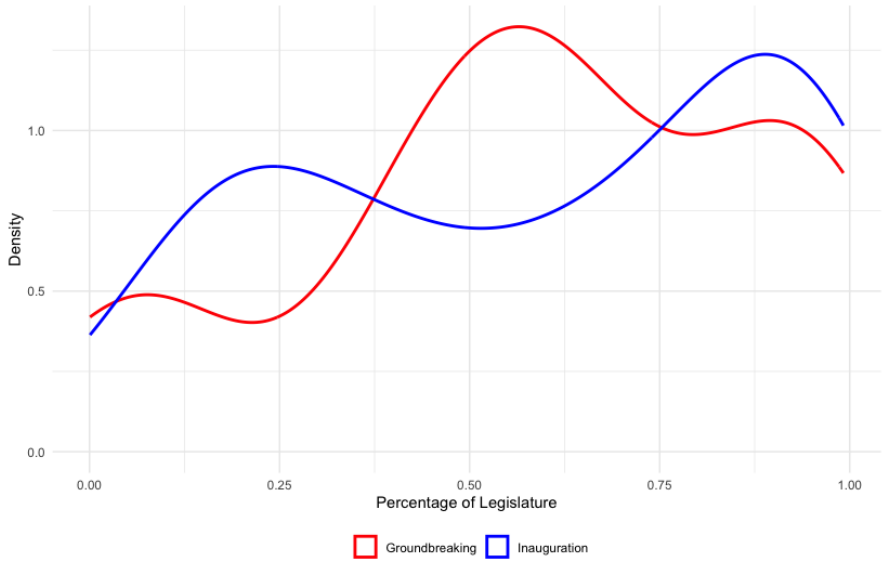


Figure A.5.2: Electoral Infrastructure Construction Cycle: groundbreaking and inauguration.

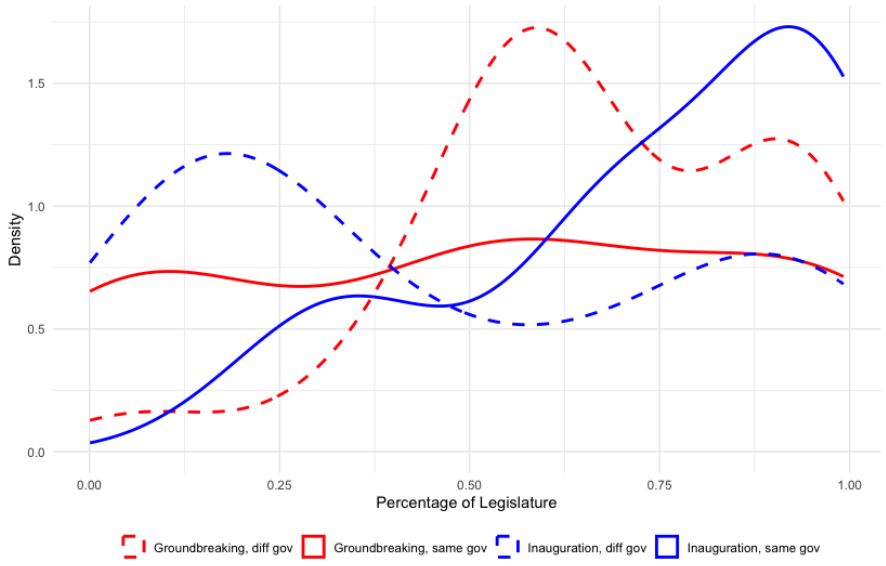


Figure A.5.3: Electoral Infrastructure Construction Cycle: coincidence of government at groundbreaking and inauguration.

A.6 Spanish Electoral Archive: filling the gaps.

Each CCAA_election dataset includes a different set of variables. All include the number of votes per candidature[23] Except for Aragón 2007, all included the census population. Almost every dataset includes blank and null votes. A large majority covers total votes to candidates, while a lower percentage offer a raw variable for valid votes and abstentions.

In order to fill missing values for these variables, I consider the following equations relating each of these variables. The only information required to conduct these calculations are the number of votes for each candidate, null and blank votes, and census population.

$$\text{Total Candidates Votes}_{it} = (\sum_{c=m}^{n} \text{Votes}_c)_{it} \tag{2}$$

$$\text{Valid Votes}_{it} = \text{Blank}_{it} + \text{Total Candidates Votes}_{it} \tag{3}$$

$$\text{Total Votes}_{it} = \text{Null}_{it} + \text{Valid Votes}_{it} \tag{4}$$

$$\text{Census}_{it} = \text{Abstentions}_{it} + \text{Total Votes}_{it} \tag{5}$$

After some visual inspection, I realized that raw data quality may differ across CCAA and elections. From the beginning, some municipality ids were repeated in Castilla y León, as well as total votes being higher than census population, or a negative amount of abstentions. Consequently, I employ several strategies in order to check the quality of original data.

Thus, I have two sets of variables. On the one hand, the original variables included in each dataset. On the other hand, I propose my own calculations of these same variables, employing the existing votes for each candidate, blank and null votes, and census population. I test the quality of the original data by comparing the value given by the original dataset and my calculation, when both are available. For instance, the existing value of Total Candidates Votes with the addition of each candidates votes for a given municipality_election observation. I do this for two main variables: Total Candidates Votes (as the original value is offered for the largest subset of observation) and Abstentions (as this calculation concerns the census population. Their quality is measured through the precision of calculated values with the original value, as follows:

$$\text{Deviation}_{it}^{\text{Total Votes}} = \frac{\text{Total Votes (Calculated)}_{it} - \text{Total Votes (Raw)}_{it}}{\text{Total Votes (Raw)}_{it}} \tag{6}$$

Given three levels of deviation, 0% (perfect match), 5%, and 10%, in Table 4 I plot the percentage of observations for each CCAA_election that fit into these three levels for Total Votes and Abstentions. Additionally, I include a final column that reflects the percentage of observation with negative abstention values, as a verisimilitude variable. Negative values reflect that census population is smaller than total votes, what is incoherent by definition. Most issues

[23] Only 0.082% of observation since 2005 have missing values on candidatures.

emerge for Castilla y Leon and Canarias observations. Other issues, as Andalucía 1996, appear before 2005, what would not impact this first chapter results.

Additionally, I calculate the percentage of Total Votes over Census, and plot their histogram and the scatterplot compared to Census population. Figure A.6.1 plots them for every region except for Castilla y Leon and Canarias since 2001[24], while Figure A.6.2 does it for Castilla y Leon and Canarias. First, we can see that with a 5% level of deviation, every observation with both values available are included. Only 17 observations (0.055%) exceed the 100% participation rate, what should be impossible. No differentiated pattern is observed for calculated values compared to original data, except for a larger number of no participation reports in calculated data, what could imply that data is actually missing for those municipalities. The distribution of original values and calculated variables is pretty similar. For the second graph, the graph is considerably different, with a large set of values over 100% or below 0.5% that reflect no match. Thus I am confident about the quality of data beyond these two CCAA.

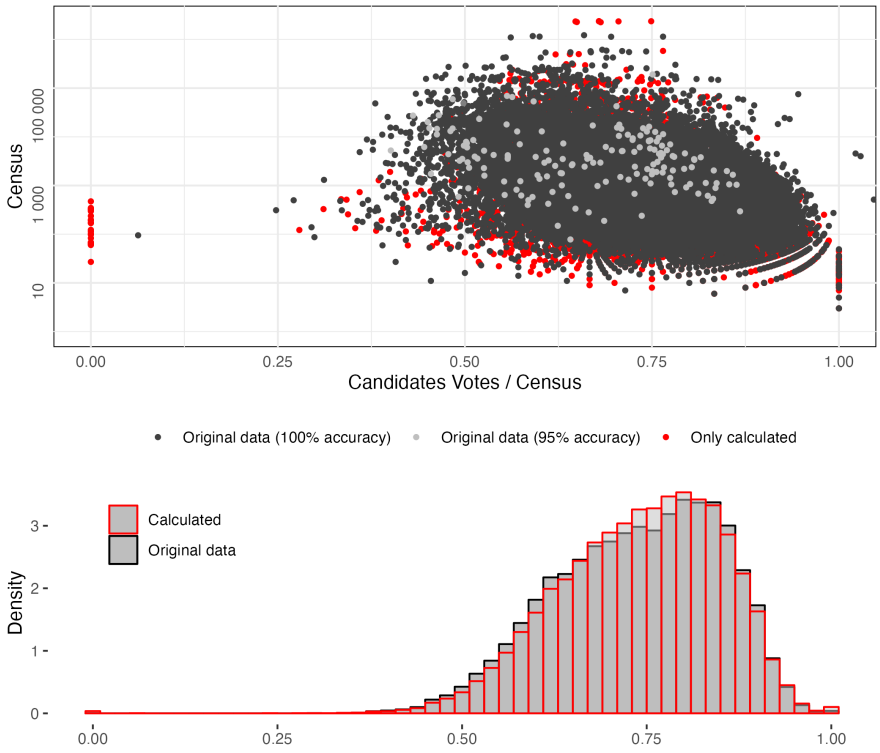


Figure A.6.1: Quality of Total Votes:

[24] I test this from 2001 onwards because healthcare competences were transferred to all regions (CCAA) in 2001. The first regional elections held under these competences took place after that date, by 2005 at the latest. However, computing the change in the incumbents' vote share for 2005 requires results from previous elections.

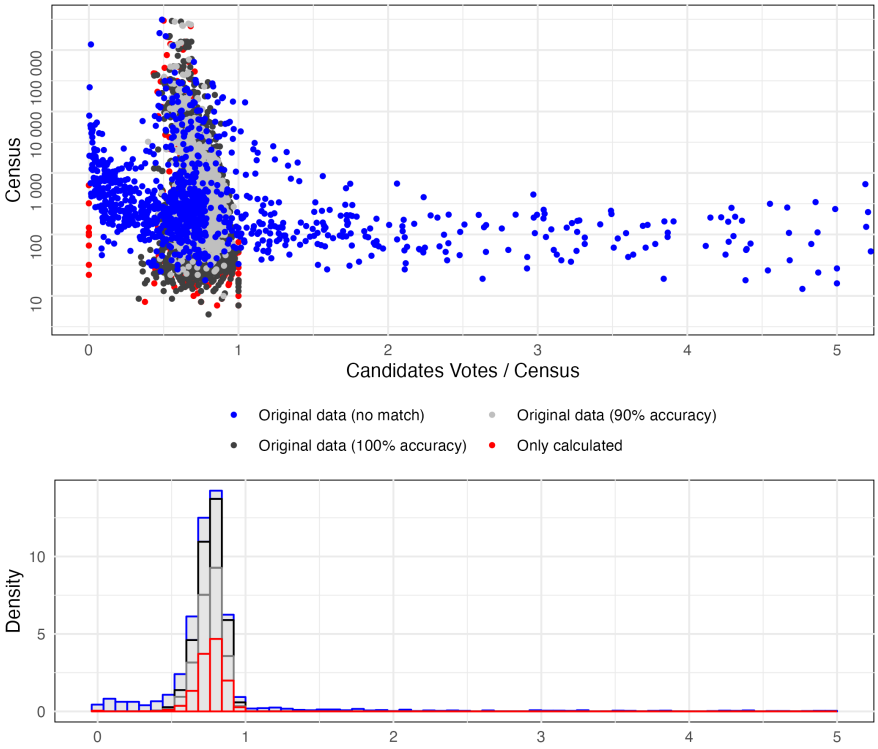

Census
10
100
1 000
10 000
100 000
0
1
2
3
4
5
Candidates Votes / Census
Original data (no match)
Original data (90% accuracy)
Original data (100% accuracy)
Only calculated
Density
0
5
10

Figure A.6.2

Table 4: Quality: Percentage of observations

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Andalucia	1982	5	780	97.05	97.31	97.31	-	-	-	0.00
Andalucia	1986	6	780	97.95	97.95	97.95	-	-	-	0.00
Andalucia	1990	6	780	88.46	97.82	98.08	-	-	-	0.00
Andalucia	1994	6	780	98.59	98.59	98.59	-	-	-	0.00
Andalucia	1996	3	780	55.90	85.90	85.90	-	-	-	3.46
Andalucia	2000	3	780	98.59	98.59	98.59	-	-	-	0.00
Andalucia	2004	3	780	98.72	98.72	98.72	-	-	-	0.00
Andalucia	2008	3	780	89.10	98.72	98.72	-	-	-	0.00
Andalucia	2012	3	780	98.85	98.85	98.85	-	-	-	0.00
Andalucia	2015	3	780	98.97	98.97	98.97	-	-	-	0.00
Andalucia	2018	12	780	99.74	99.74	99.74	-	-	-	0.00
Aragon	2007	5	731	100.00	100.00	100.00	-	-	-	0.00
Aragon	2011	5	731	-	-	-	-	-	-	0.14
Aragon	2015	5	731	-	-	-	-	-	-	0.00
Aragon	2019	5	731	-	-	-	-	-	-	0.00
Asturias	1983	5	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	1987	6	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	1991	5	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	1995	5	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	1999	6	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	2003	5	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	2007	5	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Asturias	2011	5	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	2012	3	78	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Asturias	2015	5	78	-	-	-	-	-	-	0.00
Asturias	2019	5	78	-	-	-	-	-	-	0.00
Baleares	1983	5	67	97.01	97.01	97.01	97.01	97.01	97.01	0.00
Baleares	1987	6	67	98.51	98.51	98.51	98.51	98.51	98.51	0.00
Baleares	1991	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	1995	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	1999	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	2003	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	2007	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	2011	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	2015	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Baleares	2019	5	67	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Canarias	1991	5	88	1.14	17.05	26.14	-	-	-	12.50
Canarias	1995	5	88	1.14	13.64	27.27	-	-	-	7.95
Canarias	1999	6	88	40.91	97.73	98.86	-	-	-	0.00
Canarias	2003	5	88	98.86	98.86	98.86	-	-	-	0.00
Canarias	2007	5	88	14.77	27.27	42.05	-	-	-	3.41
Canarias	2011	5	88	3.41	15.91	25.00	-	-	-	11.36
Canarias	2015	5	88	100.00	100.00	100.00	-	-	-	0.00
Canarias	2019	5	88	-	-	-	-	-	-	0.00

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Cantabria	1983	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	1987	6	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	1991	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	1995	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	1999	6	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	2003	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	2007	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	2011	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	2015	5	102	100.00	100.00	100.00	-	-	-	0.00
Cantabria	2019	5	102	-	-	-	-	-	-	0.00
Castilla - La Mancha	1983	5	919	98.91	98.91	98.91	98.59	98.59	98.59	0.00
Castilla - La Mancha	1987	6	919	99.24	99.24	99.24	99.02	99.02	99.02	0.00
Castilla - La Mancha	1991	5	919	99.56	99.56	99.56	99.24	99.24	99.24	0.00
Castilla - La Mancha	1995	5	919	99.46	99.46	99.46	99.46	99.46	99.46	0.00
Castilla - La Mancha	1999	6	919	99.67	99.67	99.67	99.46	99.46	99.46	0.00
Castilla - La Mancha	2003	5	919	100.00	100.00	100.00	99.46	99.46	99.46	0.00
Castilla - La Mancha	2007	5	919	100.00	100.00	100.00	99.56	99.56	99.56	0.00
Castilla - La Mancha	2011	5	919	100.00	100.00	100.00	98.80	98.80	98.80	0.00
Castilla - La Mancha	2015	5	919	99.89	99.89	99.89	-	-	-	0.00
Castilla - La Mancha	2019	5	919	-	-	-	-	-	-	0.11
Castilla y Leon	1983	5	2253	84.78	92.77	95.34	84.86	88.06	91.43	0.00
Castilla y Leon	1987	6	2253	72.26	98.05	99.20	72.21	93.52	97.25	0.04

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Castilla y Leon	1991	5	2253	87.22	99.11	99.29	87.13	96.89	98.31	0.09
Castilla y Leon	1995	5	2253	81.76	98.27	99.02	81.67	92.68	96.01	0.04
Castilla y Leon	1999	6	2253	76.48	97.96	99.07	76.48	91.12	95.43	0.04
Castilla y Leon	2003	5	2253	63.25	86.64	92.01	63.07	69.95	77.19	1.55
Castilla y Leon	2007	5	2253	67.42	87.48	93.03	67.20	76.83	81.94	1.42
Castilla y Leon	2011	5	2253	53.00	71.82	76.52	52.91	60.32	66.13	9.68
Castilla y Leon	2015	5	2253	77.63	91.48	94.10	77.45	84.33	88.10	1.46
Castilla y Leon	2019	5	2253	-	-	-	-	-	-	0.00
Catalunya	1980	3	948	98.63	98.63	98.63	98.63	98.63	98.63	0.00
Catalunya	1984	4	948	98.95	99.05	99.05	98.95	99.05	99.05	0.00
Catalunya	1988	5	948	99.16	99.16	99.16	98.73	98.95	98.95	0.00
Catalunya	1992	3	948	99.37	99.37	99.37	99.37	99.37	99.37	0.00
Catalunya	1995	11	948	99.58	99.58	99.58	99.58	99.58	99.58	0.00
Catalunya	1999	10	948	99.79	99.79	99.79	99.79	99.79	99.79	0.00
Catalunya	2003	11	948	99.79	99.79	99.79	99.79	99.79	99.79	0.00
Catalunya	2006	11	948	99.79	99.79	99.79	99.79	99.79	99.79	0.00
Catalunya	2010	11	948	99.79	99.79	99.79	99.79	99.79	99.79	0.00
Catalunya	2012	11	948	99.89	99.89	99.89	99.89	99.89	99.89	0.00
Catalunya	2015	9	948	99.89	99.89	99.89	99.89	99.89	99.89	0.00
Catalunya	2017	12	948	99.79	99.89	99.89	96.10	96.52	96.84	1.27
Catalunya	2021	2	948	93.57	99.89	99.89	93.57	99.89	99.89	0.00
Ceuta	1995	5	1	100.00	100.00	100.00	-	-	-	0.00

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Ceuta	1999	6	1	-	100.00	100.00	-	-	-	0.00
Ceuta	2003	5	1	100.00	100.00	100.00	-	-	-	0.00
Ceuta	2007	5	1	100.00	100.00	100.00	-	-	-	0.00
Ceuta	2011	5	1	100.00	100.00	100.00	-	-	-	0.00
Ceuta	2015	5	1	-	100.00	100.00	-	-	-	0.00
Ceuta	2019	5	1	100.00	100.00	100.00	-	-	-	0.00
Comunitat Valenciana	1983	5	543	98.16	98.34	98.34	96.69	98.16	98.16	0.00
Comunitat Valenciana	1987	5	543	-	-	-	98.71	98.90	98.90	0.00
Comunitat Valenciana	1991	5	543	99.26	99.26	99.26	-	-	-	0.00
Comunitat Valenciana	1995	5	543	96.50	98.34	98.34	96.32	97.97	97.97	0.18
Comunitat Valenciana	1999	6	543	99.45	99.63	99.63	99.45	99.63	99.63	0.00
Comunitat Valenciana	2003	5	543	97.61	97.61	97.61	97.42	97.61	97.61	0.00
Comunitat Valenciana	2007	5	543	99.82	99.82	99.82	99.45	99.45	99.45	0.18
Comunitat Valenciana	2011	5	543	99.82	99.82	99.82	99.82	99.82	99.82	0.00
Comunitat Valenciana	2015	5	543	99.82	99.82	99.82	99.45	99.45	99.63	0.00
Comunitat Valenciana	2019	4	543	-	-	-	-	-	-	0.00
Extremadura	1983	5	388	97.94	97.94	97.94	97.94	97.94	97.94	0.00
Extremadura	1987	6	388	97.94	97.94	97.94	97.94	97.94	97.94	0.00
Extremadura	1991	5	388	97.94	97.94	97.94	9.28	83.25	92.27	0.00
Extremadura	1995	5	388	98.45	98.45	98.45	98.45	98.45	98.45	0.00
Extremadura	1999	6	388	98.45	98.45	98.45	7.22	59.54	77.32	0.00
Extremadura	2003	5	388	98.71	98.71	98.71	98.71	98.71	98.71	0.00

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Extremadura	2007	5	388	98.71	98.71	98.71	97.68	98.71	98.71	0.26
Extremadura	2011	5	388	99.23	99.23	99.23	99.23	99.23	99.23	0.00
Extremadura	2015	5	388	99.74	99.74	99.74	92.78	93.04	93.30	0.52
Extremadura	2019	5	388	-	-	-	-	-	-	0.00
Galicia	2001	10	317	92.43	99.37	99.37	99.37	99.37	99.37	0.00
Galicia	2005	6	317	99.37	99.37	99.37	99.37	99.37	99.37	0.00
Galicia	2009	3	317	99.37	99.37	99.37	99.37	99.37	99.37	0.00
Galicia	2012	10	317	99.37	99.37	99.37	98.74	98.74	99.05	0.00
Galicia	2016	9	317	99.05	99.05	99.05	53.00	99.05	99.05	0.00
Galicia	2020	7	317	-	-	-	-	-	-	0.00
La Rioja	1983	5	174	-	-	-	-	-	-	1.72
La Rioja	1987	6	174	-	-	-	-	-	-	0.00
La Rioja	1991	6	174	-	-	-	-	-	-	0.00
La Rioja	1995	5	174	-	-	-	-	-	-	0.00
La Rioja	1999	6	174	-	-	-	-	-	-	0.57
La Rioja	2003	6	174	-	-	-	-	-	-	0.00
La Rioja	2007	5	174	-	-	-	-	-	-	0.00
La Rioja	2011	5	174	-	-	-	-	-	-	0.00
La Rioja	2015	5	174	-	-	-	-	-	-	0.00
La Rioja	2019	5	174	-	-	-	-	-	-	0.00
Madrid	1995	5	179	5.59	98.88	99.44	-	-	-	0.00
Madrid	1999	6	179	-	-	-	-	-	-	0.00

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Madrid	2003	5	179	-	-	-	-	-	-	0.00
Madrid	2003	10	179	-	-	-	-	-	-	0.00
Madrid	2007	4	179	-	-	-	-	-	-	0.00
Madrid	2011	5	179	-	-	-	-	-	-	0.00
Madrid	2015	5	179	-	-	-	-	-	-	0.00
Madrid	2019	5	179	-	-	-	-	-	-	0.00
Madrid	2021	5	179	-	-	-	-	-	-	0.00
Melilla	1995	5	1	100.00	100.00	100.00	-	-	-	0.00
Melilla	1999	6	1	-	100.00	100.00	-	-	-	0.00
Melilla	2003	5	1	-	-	-	-	-	-	0.00
Melilla	2007	5	1	100.00	100.00	100.00	-	-	-	0.00
Melilla	2011	5	1	100.00	100.00	100.00	-	-	-	0.00
Melilla	2015	5	1	-	100.00	100.00	-	-	-	0.00
Melilla	2019	5	1	100.00	100.00	100.00	-	-	-	0.00
Murcia	2003	6	45	100.00	100.00	100.00	-	-	-	0.00
Murcia	2007	5	45	-	-	-	-	-	-	0.00
Murcia	2011	5	45	-	-	-	-	-	-	0.00
Murcia	2015	5	45	-	-	-	-	-	-	0.00
Murcia	2019	5	45	100.00	100.00	100.00	-	-	-	0.00
Navarra	1979	4	272	97.06	97.06	97.06	97.06	97.06	97.06	0.00
Navarra	1983	5	272	97.06	97.06	97.06	97.06	97.06	97.06	0.00
Navarra	1987	6	272	97.43	97.43	97.43	97.43	97.43	97.43	0.00

Table 4: Quality: Percentage of observations *(continued)*

				Accuracy: Candidates Votes			Accuracy: Abstentions			
CCAA	Year	Month	Total	100%	95%	90%	100%	95%	90%	Abst. < 0
Navarra	1991	5	272	98.16	98.16	98.16	97.79	97.79	97.79	0.00
Navarra	1995	5	272	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Navarra	1999	5	272	99.63	99.63	99.63	98.90	98.90	98.90	0.37
Navarra	2003	5	272	-	-	-	-	-	-	0.00
Navarra	2007	5	272	-	-	-	-	-	-	0.00
Navarra	2011	5	272	-	-	-	81.99	95.96	97.06	0.00
Navarra	2015	5	272	-	-	-	-	-	-	0.00
Navarra	2019	5	272	-	-	-	-	-	-	0.00
Pais Vasco	1980	3	251	90.84	90.84	90.84	74.10	90.04	90.44	0.00
Pais Vasco	1984	2	251	93.63	93.63	93.63	91.24	93.63	93.63	0.00
Pais Vasco	1986	11	251	86.85	90.04	90.04	85.26	90.04	90.04	0.00
Pais Vasco	1990	10	251	97.61	97.61	97.61	96.02	97.61	97.61	0.00
Pais Vasco	1994	10	251	98.80	98.80	98.80	98.80	98.80	98.80	0.00
Pais Vasco	1998	10	251	99.60	99.60	99.60	99.60	99.60	99.60	0.00
Pais Vasco	2001	5	251	99.60	99.60	99.60	99.60	99.60	99.60	0.00
Pais Vasco	2005	4	251	99.60	99.60	99.60	99.60	99.60	99.60	0.00
Pais Vasco	2009	3	251	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Pais Vasco	2012	10	251	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Pais Vasco	2016	9	251	100.00	100.00	100.00	100.00	100.00	100.00	0.00
Pais Vasco	2020	7	251	100.00	100.00	100.00	100.00	100.00	100.00	0.00

A.7 Robustness checks

A.7.1 Change in vote share for the regional incumbent at the election after inauguration

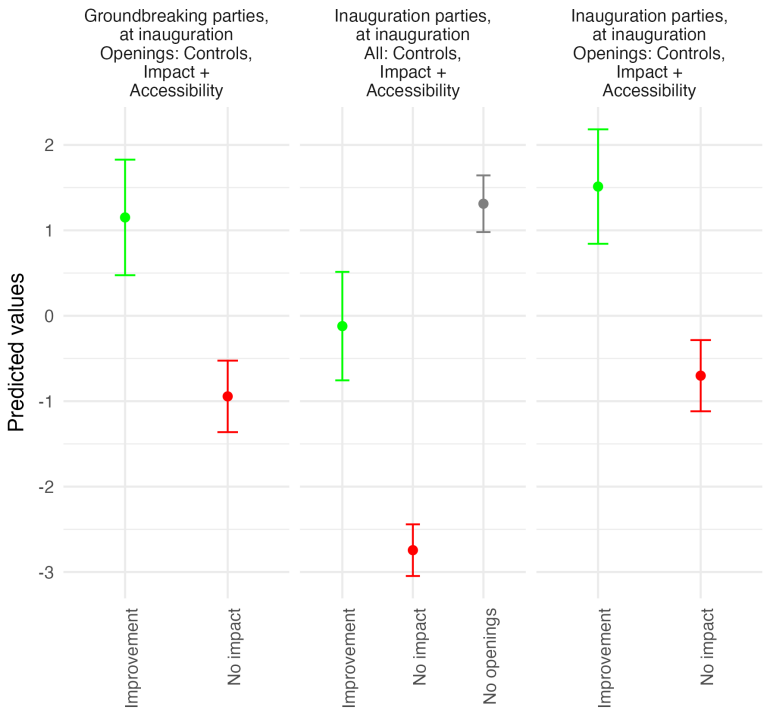


Figure A.7.1: Inauguration

A.7.2 New hospitals' impact: both openings and relocations.

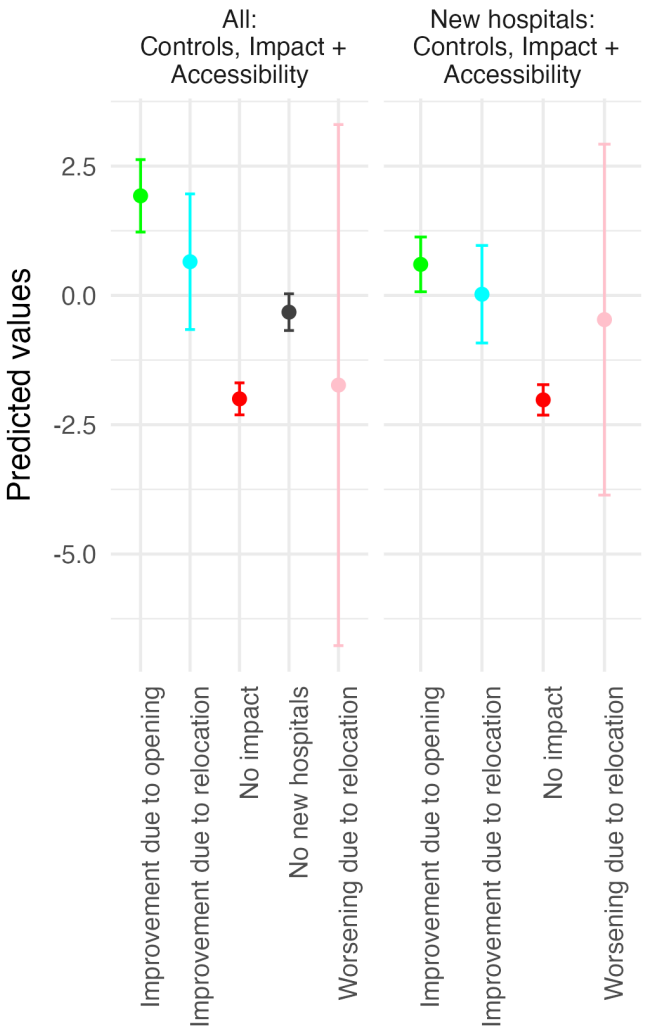


Figure A.7.2: New hospitals: Openings + Relocations

A.7.3 Alternative measures of accessibility

(a) Within-region standardized distances

(b) Absolute distances (km)

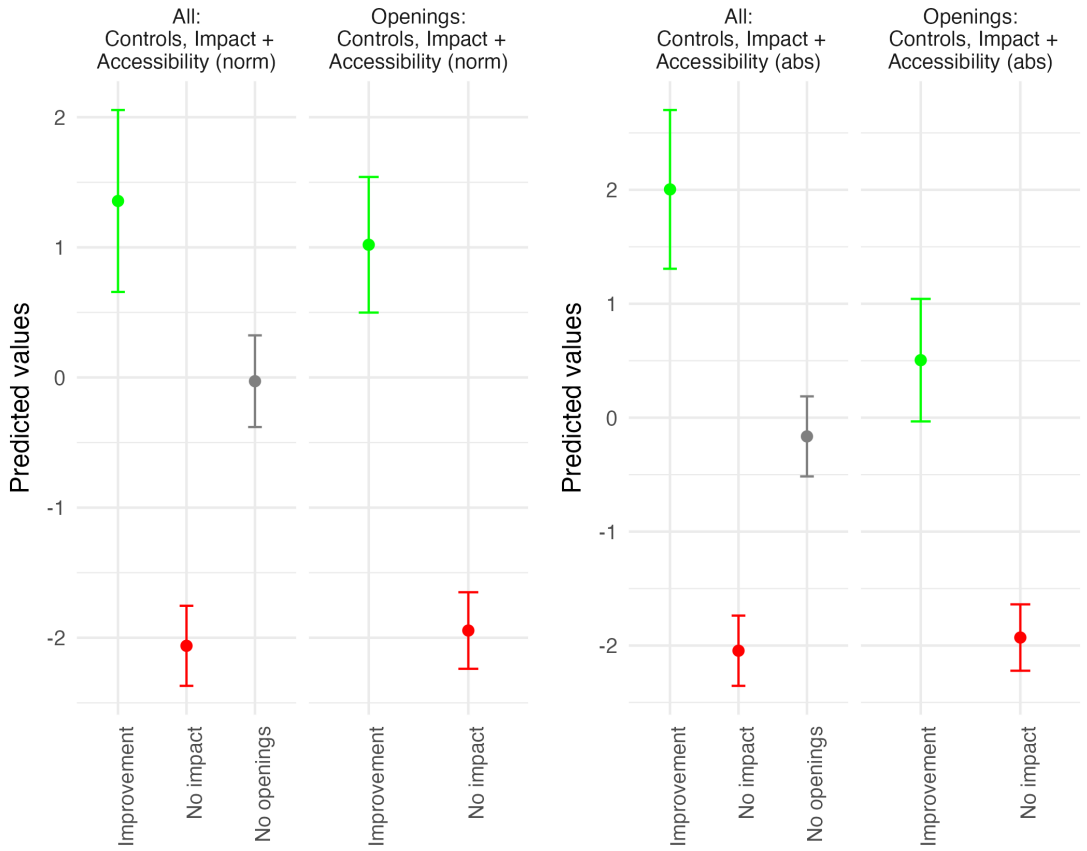


Figure A.7.3

A.7.4 Models without controls and term fixed effects

(a) Without controls

(b) Without term fixed effects

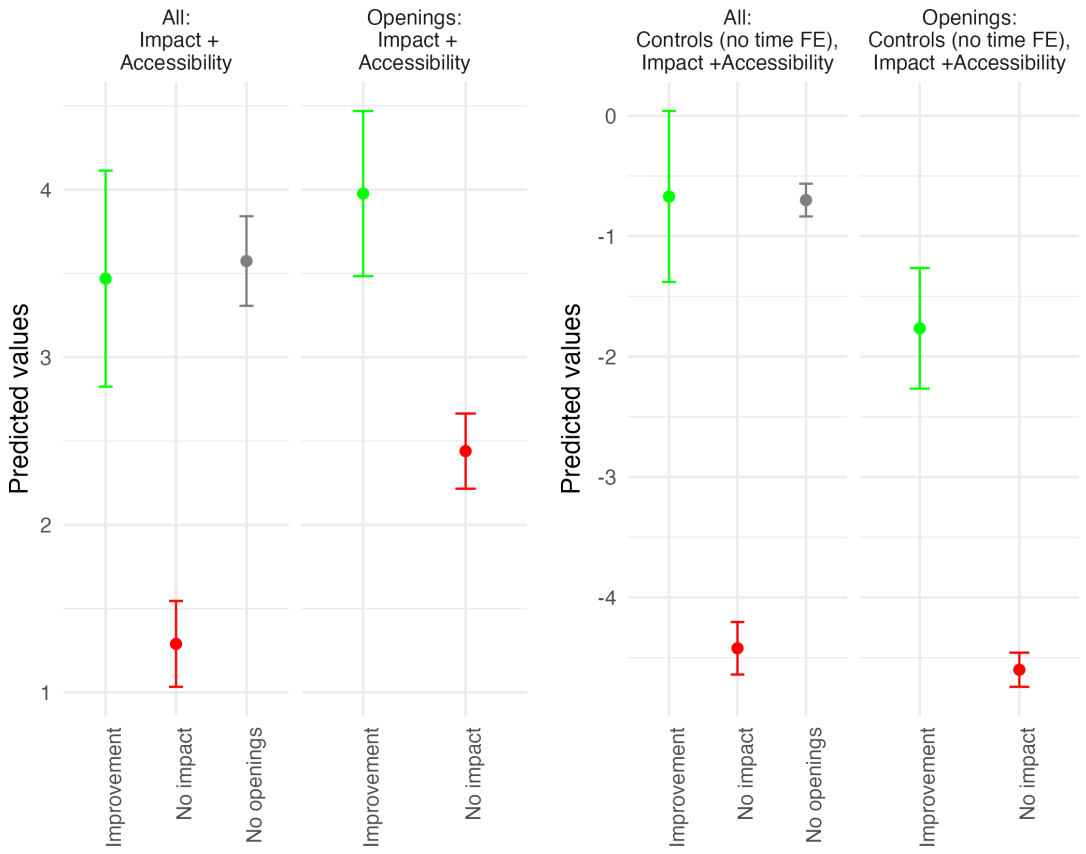


Figure A.7.4